PHYSICS

1) The energy E of an oscillating body in simple harmonic motion depends on its mass m, frequency n and amplitude A as $E = k(m)^x(n)^y(A)^z$. Find the value of (2x + y + z) : [K is a dimensionless constant]

(1) 3
(2) 4
(3) 5
(4) 6

2) The maximum and minimum resultant of two forces acting at a point are 10N and 6N respectively. If each force is increased by 3N, find the resultant of new forces when acting at a point at an angle of 90° with each-other :-

(1) $\sqrt{146}\,N$
(2) 11N
(3) $\sqrt{70}\,N$
(4) 8N

3) The pitch of a screw gauge is 0.5 mm and there are 50 divisions on its circular scale. When screw is closed and plane face of screw is in contact with plane face of stud, on the frame, it was observed that zero of the circular scale is above the pitch line by 5 divisions. The zero error in instrument is :

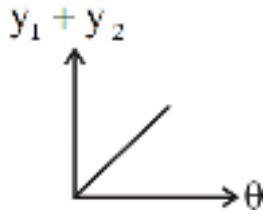
(1) 0.025 mm
(2) –0.25 mm
(3) 0.05 mm
(4) –0.05 mm

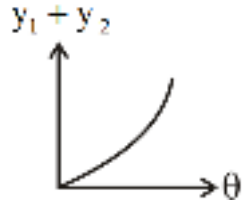
4) Given that velocity $\vec{v} = t\hat{i} + t^2\hat{j}$ where t is time. Find rate of change of speed :

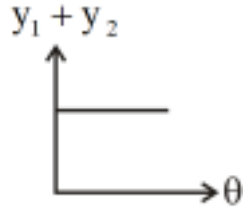
(1) $\frac{1+2t^2}{\sqrt{1+t^2}}$
(2) $\frac{1+t^2}{\sqrt{1+t^2}}$
(3) $\frac{t^2}{\sqrt{1+t^2}}$
(4) $\frac{\sqrt{1+t^2}}{1+2t^2}$

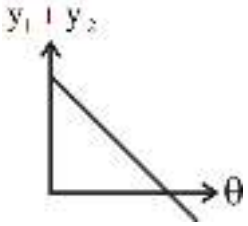
5) A ball is thrown at two different angles with the same speed 'u' and from the same point and it has the same range in both the cases. If y_1 and y_2 are heights attained in the two cases then choose

the correct graph showing relationship between ($y_1 + y_2$) and angle of projection (θ) :

(1) 

(2) 

(3) 

(4) 

6) A rocket is fired vertically from the earth with an acceleration of 2g, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. After firing the rocket find the minimum coefficient of friction μ_{min} between the mass and the inclined surface such that the mass does not move is :

(1) tan 2θ

(2) tanθ

(3) 3 tanθ

(4) 2 tanθ

7) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) :- An electric fan continues to rotate for some time after the current is switched off.
Reason (R) :- Fan continuous to rotate due to inertia of rotational motion.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) (A) is correct but (R) is not correct.

(2) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct but (R) is NOT the correct explanation of (A).

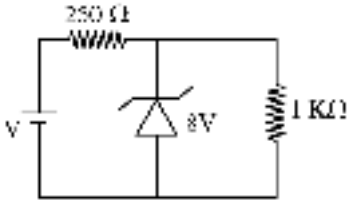
8) A body of mass m travels in a straight line with velocity $u = \alpha \cdot x^{3/2}$ where α is constant. work done by the net force during its displacement from x = 0 to $x = x_0$ is :-

(1) $\frac{m\alpha^2.x_0^3}{2}$

(2) $m\alpha^2x_0^3$

(3) $2m\alpha^2x_0^3$

(4) None of these



9) Find minimum value of V so given circuit work as voltage regulation :-

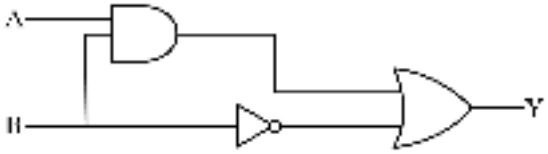
(1) 8 V

(2) 10 V

(3) 12 V

(4) 16 V

10) Truth table for given combination of logic gates will be :-



(1)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(3)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

11) **Assertion:** If YDSE setup is immersed in water, fringe width will decrease
Reason: Wavelength decreases in denser medium.

(1) Assertion is true, Reason is true; Reason is correct explanation for Assertion.

(2) Assertion is true, Reason is true; Reason is not correct explanation for Assertion.

(3) Assertion is true, Reason is false.

(4) Assertion is false, Reason is true.

12)

When a person is situated at a distance 'd' from a point source, observed intensity is I_0. When person shift 50m towards source, intensity becomes 9 times of initial value. Find d.

(1) 50 m

(2) 75 m

(3) 100 m

(4) 20 m

13) **Statement-1 :** If intensity of monochromatic light is increased then number of photon passing through a given area increases per second. While energy of each photon remains same.
Statement-2 : In photon-matter collision total linear momentum and total energy remains conserved.

(1) (1) & (2) both are correct

(2) (1) is correct but (2) is not correct

(3) (1) is not correct but (2) is correct

(4) Both (1) and (2) are incorrect

14) An alpha particle and a proton are projected in a transverse magnetic field. If ratio of radius of their path is 1 : 2 then ratio of their de-Broglie wavelengths will be :-

(1) 1 : 4

(2) 4 : 1

(3) 1 : 1

(4) 2 : 1

15) For a hydrogen atom, when electron jump from n = 2 to n = 1, then wave length of radiation emitted found to be λ_0.Then, for which transition of electron of a 'He' atom, the wavelength of radiation emitted will be equal to $'\lambda_0'$.

(1) n = 3 to n = 2

(2) n = 4 to n = 2

(3) n = 4 to n = 3

(4) n = 8 to n = 4

16) When a ${}_4Be^9$ atom is bombarded with α-particles. One of the product of nuclear transmutation is ${}_6C^{12}$. The other is :-

(1) ${}_{-1}e^{0}$

(2) ${}_{1}H^{1}$

(3) ${}_{1}D^{2}$

(4) ${}_{0}n^{1}$

17) In a compound microscope, the focal length of the objective and the eye lens are 2 cm and 5 cm respectively. An object is placed at 3 cm before the objective and image is formed at the least distance of distinct vision, then the distance between two lenses will be :-

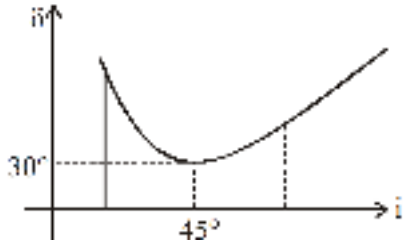
(1) 10.16 cm

(2) 18.67 cm

(3) 23.00 cm

(4) 52.00 cm

18) Variation of angle of deviation δ versus angle of incidence i for a prism is given in the figure. The value of refractive index of prism :-



(1) $\sqrt{3}$

(2) $\sqrt{2}$

(3) $\sqrt{\frac{3}{2}}$

(4) $\frac{2}{\sqrt{3}}$

19) The focal length of a convex lens is f and the distance of an object from the principal focus is x. The ratio of the size of the real image to the size of the object is

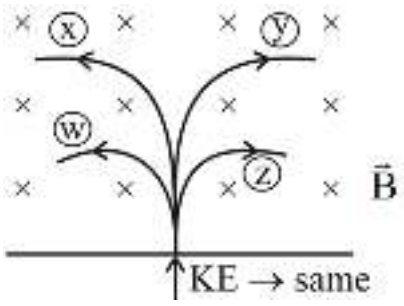
(1) f/x

(2) x/f

(3) $\frac{f+x}{f}$

(4) $\frac{f}{f+x}$

20) Identify (–q, m), (–3q, 4m), (+q, m) and (+3q, 4m).



	Column-1		Column-2
(a)	Particle (–q, m)	p.	w
(b)	Particle (–3q, 4m)	q.	x
(c)	Particle (+q, m)	r.	y
(d)	Particle (+3q, 4m)	s.	z

(1) a-q, b-p, c-r, d-s

(2) a-r, b-s, c-q, d-p

(3) a-p, b-q, c-s, d-r

(4) a-s, b-r, c-p, d-q

21) A wire of length carrying a current of i is bent to form a square coil of 2 turns then its magnetic moment will be :-

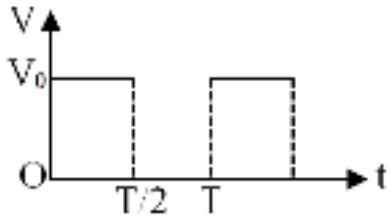
(1) $\frac{i\ell^2}{32\pi}$

(2) $\frac{i\ell^2}{64\pi}$

(3) $\frac{i\ell^2}{32}$

(4) $\frac{i\ell^2}{64}$

22) The r.m.s. value in time T of potential difference V shown in the figure is :-



(1) $\frac{V_0}{\sqrt{3}}$

(2) V_0

(3) $\frac{V_0}{\sqrt{2}}$

(4) $\frac{V_0}{2}$

23) **Statement-I :** When the sun shines on our hand we feel a radiation pressure.
Statement-II : When we are exposed to sun shine, amount of momentum transferred by light is very small.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

24) Two satellites of masses m_1 and m_2 ($m_1 > m_2$) are revolving around the earth in circular orbits of radius r_1 and r_2 ($r_1 > r_2$) respectively. Which of the following relation is true regarding their speeds V_1 and V_2 :

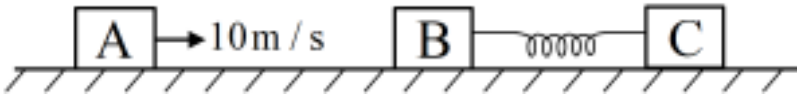
(1) $V_1 = V_2$

(2) $V_1 < V_2$

(3) $V_1 > V_2$

(4) $\frac{V_1}{r_1} = \frac{V_2}{r_2}$

25) Three identical blocks A, B and C, each of mass 2 kg are kept on a smooth horizontal surface. Block A is given a horizontal velocity 10 m/s towards B (as shown in figure). Block B and C are attached with a spring. If block A collide with block B elastically, then find change in velocity of center of mass of system A + B + C if A collides with B elastically.



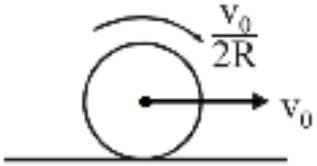
(1) 5 m/s

(2) 10 m/s

(3) 20 m/s

(4) zero

26) Consider the solid sphere moving on horizontal surface. At some instant it has linear velocity v and angular velocity about center of mass $\frac{v_0}{2R}$. The translational velocity after the sphere starts pure rolling :-



(1) $\frac{5v_0}{7}$

(2) $\frac{6v_0}{7}$

(3) $\frac{3v_0}{4}$

(4) $\frac{3v_0}{5}$

27) A particle is revolving around a circle of radius 1 m. Angular velocity of particle $\omega = 3t^2 - 2t$ rad/sec changes with time (t is in sec).

	List-I		**List-II** **Values are in appropriate SI units**

(P)	Angular acceleration at t = 1 sec	(1)	4
(Q)	Tangential acceleration at t = 1 sec	(2)	3
(R)	Centripetal acceleration at t = 1 sec	(3)	1
(S)	Rate of change of angular velocity at t = 2 sec	(4)	10

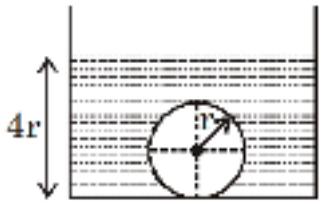
(1) P → 1;Q → 1;R → 3;S → 4

(2) P → 2;Q → 4;R → 1;S → 2

(3) P → 3;Q → 1;R → 4;S → 2

(4) P → 4;Q → 1;R → 2;S → 3

28) A small spherical metal ball of radius r is lying at the bottom of a stationary container containing liquid of density ρ as shown in the figure. P_0 is atmospheric pressure. Force exerted by liquid on the upper hemispherical part of the sphere is :



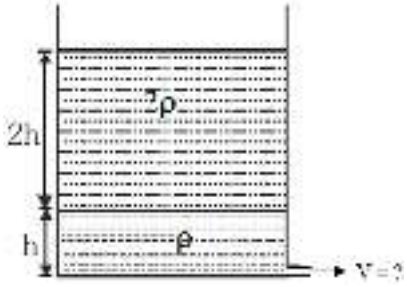
(1) $\frac{\pi r^2}{3}[3P_0 + 7r\rho g]$

(2) $\frac{2\pi r^2}{3}[3P_0 + 4r\rho g]$

(3) $\frac{\pi r^2}{2}[P_0 + 7r\rho g]$

(4) $\frac{\pi r^2}{3}[2P_0 + 7r\rho g]$

29) The velocity of the liquid coming out of a small hole of vessel containing two different liquids of densities 2ρ and ρ as shown in figure is-



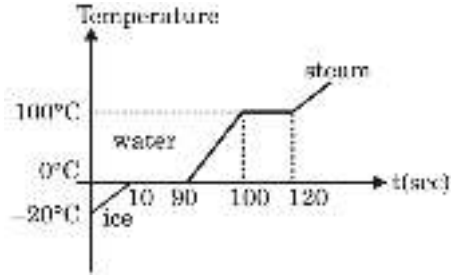
(1) $\sqrt{6gh}$

(2) $2\sqrt{gh}$

(3) $2\sqrt{2gh}$

(4) $\sqrt{10gh}$

30) A given mass m of a solid is supplied with heat continuously at a constant rate and the graph shown in the figure is plotted. If L_f and L_v are latent heat of fusion and latent heat of vaporization and S_ℓ and S_s are specific heats of liquid and solid respectively, then



(1) $L_f > L_v, S_\ell = S_s$

(2) $L_f < L_v, S_\ell > S_s$

(3) $L_f > L_v, S_\ell < S_s$

(4) $L_f = L_v, S_\ell > S_s$

31) When q_1 joules of radiation is incident on a body, it reflects and transmits total of q_2 joules. Find the emissivity of the body.

(1) $\frac{q_1 + q_2}{q_1}$

(2) $\frac{q_1 - q_2}{q_1}$

(3) $\frac{q_1}{q_1 + q_2}$

(4) $\frac{q_1}{q_1 - q_2}$

32) A spherical solid black body of radius 'r' radiates power 'H' and its rate of cooling is 'C'. If density is constant, then which of the following is/are true ?

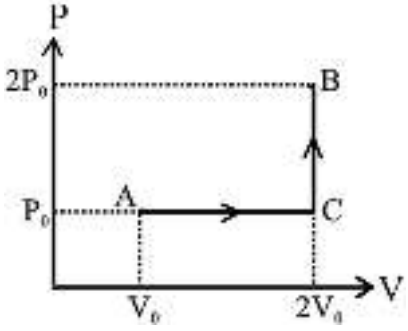
(1) $H \propto r$ and $C \propto r^2$

(2) $H \propto r^2$ and $C \propto 1/r$

(3) $H \propto r$ and $C \propto 1/r^2$

(4) $H \propto r^2$ and $C \propto r^2$

33) One mole of monoatomic gas is brought from state A to state B and temperature at A is T_0. Find the heat absorbed along the path ACB.



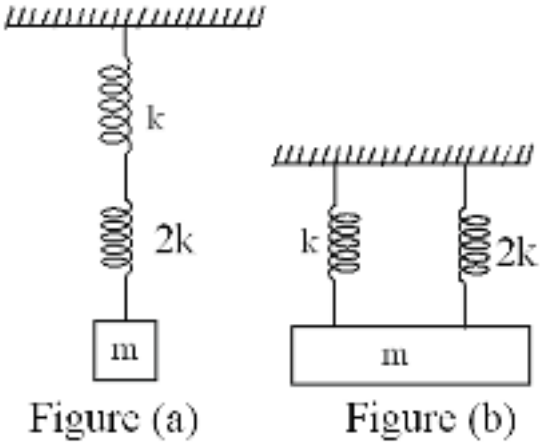
(1) $\frac{11}{2}RT_0$

(2) $\frac{9}{2}RT_0$

(3) $\frac{7}{2}RT_0$

(4) $\frac{5}{2}RT_0$

34) As per given figures, two springs of spring constants K and 2K are connected to mass m. If the period of oscillation in figure (a) is 3s, then the period of oscillation in figure (b) will be $\sqrt{x}$ s. The value of x is_____.



(1) 4

(2) 3

(3) 2

(4) 1

35) For a particle executing simple harmonic motion, which of the following statement is not correct ?

(1) the total energy of particle always remains the same

(2) the restoring force is always directed towards a fix point

(3) the restoring force is maximum at the extreme positions

(4) the acceleration of particle is maximum at the equilibrium position

36) Three tuning forks of frequencies 200, 203 and 207 Hz are sounded together. Find out the beat frequency (in Hz).

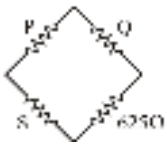
(1) 3

(2) 4

(3) 8

(4) 12

37) A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is :-



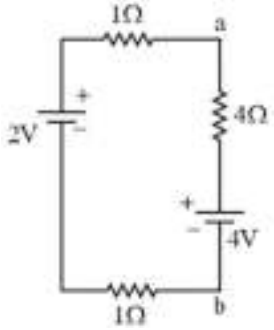
(1) 625 Ω

(2) 650 Ω

(3) 676 Ω

(4) 600 Ω

38) For the circuit shown in figure, the potential difference between points a and b is :



(1) $\frac{8}{3}$ V

(2) $\frac{3}{8}$ V

(3) 2 V

(4) 3.2 V

39) Three capacitors each of capacitance C and of breakdown voltage V are joined in series. The capacitance and breakdown voltage of the combination will be :-

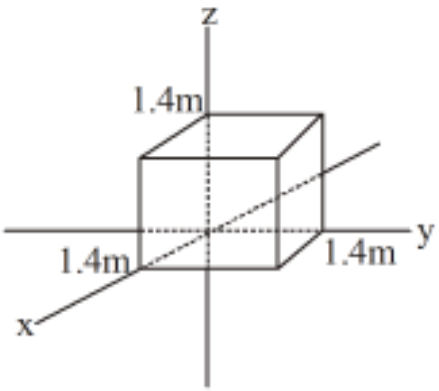
(1) 3C, 3V

(2) $\frac{C}{3}, \frac{V}{3}$

(3) $3C, \frac{V}{3}$

(4) $\frac{C}{3}, 3V$

40) The cube shown below has edge lengths of 1.4 m and is placed such that the hidden corner is at the origin of the coordinate system. If $\vec{E} = (3y)\hat{j}$, where $\vec{E}$ is in units of N/C and y is coordinate in



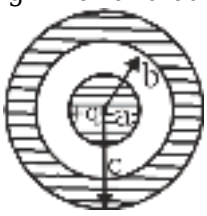
meters, what is $\phi_{net} = \oint \vec{E} \cdot \vec{dA}$

(1) 5.88 N-m^2/C

(2) 0

(3) 8.23 N-m^2/C

(4) 4.2 N-m^2/C

41) A non conducting sphere of radius 'a' has a net charge '+q' uniformly distributed throughout its volume. A hollow spherical conductor having inner and outer, radii 'b' and 'c' and net charge '–q' is concentric with the sphere (see the figure) 

Read the following statements

(i) The electric field at a distance r from the centre of the sphere = $\frac{1}{4\pi\varepsilon_0}\frac{qr}{a^3}$ for r < a
(ii) The electric field at distance r from the centre of the sphere for a < r < b is zero
(iii) The electric field at distance r from the centre of the sphere for b < r < c is zero
(iv) The charge on the inner surface of the hollow sphere is –q
(v) The charge on the outer surface of the hollow sphere is +q Which of the above statements are true ?

(1) (i) , (ii) and (v)

(2) (i) , (iii) and (iv)

(3) (ii) , (iii) and (iv)

(4) (ii) , (iii) and (v)

42) A new system of units is proposed in which unit of mass is α kg, unit of length is β m and unit of time is γ s. What will be value of 5J in this new system :

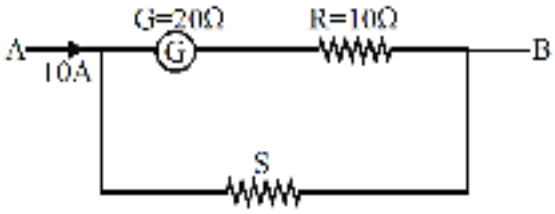
(1) $5\alpha\beta^2\gamma^{-2}$

(2) $5\alpha^{-1}\beta^{-2}\gamma^2$

(3) $5\alpha^{-2}\beta^{-1}\gamma^{-2}$

(4) $5\alpha^{-1}\beta^2\gamma^2$

43) Full scale deflection current for galvanometer is 1 mA. What should be the value of shunt resistance (approximately) so that galvanometer shows half scale deflection for the circuit shown.

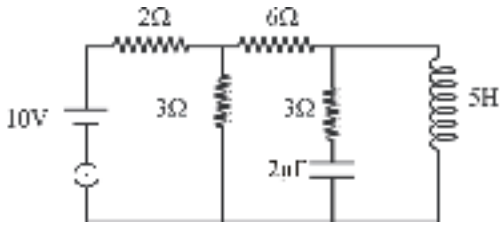


(1) 1.5 mΩ

(2) 3mΩ

(3) 10 mΩ

(4) 15 mΩ

44) 

Calculate the current flow from battery, after long time of closing the key:-

(1) 2.5 Amp

(2) 5 Amp

(3) 7 Amp

(4) 3 Amp

45) An electric charge + q unit moves with velocity $\vec{V} = 3\hat{i} + 4\hat{j} + \hat{k}$ unit, in an electromagnetic field given by: $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ unit and $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$ unit. If the y-component of the force experienced by + q is βq unit then β is :

(1) 2

(2) 10

(3) 7

(4) 11

CHEMISTRY

1) Which of the following order is **CORRECT** for their indicated properties ?
(i) $\hat{HCH}$ bond angle : $CH_3F < CH_2F_2$
(ii) Bond angle : $NH_3 < PH_3 < AsH_3$
(iii) % s-character in the orbital containing lone pair : $PH_3 < PF_3$
(iv) % s-character in E – H bonds (E = central atom) : $NH_3 < AsH_3$

(1) (i) only

(2) (i), (ii) only

(3) (ii), (iii) only

(4) (i), (iv) only

2) Which of the following has dipole-dipole interaction between the species possessing permanent dipole ?
(i) Liquid-NH_3 (ii) Liquid-He
(iii) Solid–ICl (iv) Liquid-Br_2

(1) i, ii
(2) ii, iv
(3) i, iii
(4) i, iv

3) Find the number of molecules in which p–p sideways overlapping is present.
N_2 , O_2 , HCl , Cl_2 , HF, S_8, F_2

(1) 2
(2) 3
(3) 4
(4) 5

4) Total number of paramagnetic species in which unpaired electron(s) is/are present in π* M.O.
O_2^+; O_2^-; O_2; N_2^-; N_2^+ ; N_2^{2-}; B_2

(1) 4
(2) 5
(3) 6
(4) 7

5) Which of the following statement are **CORRECT** ?
(i) In $Na_2S_4O_6$ sum of the oxidation state of four sulphur atoms is 10
(ii) $K_2Cr_2O_7$ is coloured due to charge transfer spectrum.
(iii) $[M(AA)_3]$ has 1 optically inactive isomer
(iv) E.A.N. of Co is 35 in $Co_2(CO)_8$

(1) i, iii, iv
(2) iii, iv
(3) i, ii
(4) i, ii, iii

6) X = Number of compounds which are repelled by magnetic field among KO_2,$O_2[PtF_6]$, $NO[BF_4]$, AlF_3
Y = Number of species which have same bond order as CO among CN^-, NO^+, C_2^{2-} , O_2^{2-}.
Find the value of (X + Y)

(1) 3
(2) 6
(3) 5
(4) 4

7)

List-I (Complex ions)		List-II (Hydridisation with magnetic moment)	
(P)	$[Cr(NH_3)_6]^{3+}$	(1)	dsp^2, $\mu = 0$
(Q)	$[FeF_6]^{3-}$	(2)	d^2sp^3, $\mu = 0$
(R)	$[Co(NH_3)_6]^{3+}$	(3)	d^2sp^3, $\mu = \sqrt{15}$
(S)	$[Pt(NH_3)_2(Cl)_2]$	(4)	sp^3d^2, $\mu = \sqrt{35}$

(1) P-3 ; Q-4 ; R-2 ; S-1

(2) P-2 ; Q-4 ; R-3 ; S-1

(3) P-1 ; Q-3 ; R-4 ; S-2

(4) P-4 ; Q-3 ; R-2 ; S-1

8) Select the correct statement(s) with respect to oxides and oxoanions of transition metals.
(i) Among oxides of chromium, CrO is basic, Cr_2O_3 is amphoteric and CrO_3 is acidic.
(ii) No higher oxides of iron above Fe_2O_3 are found.
(iii) Ti,V, Cr and Mn form oxides MO and their correct increasing order of acidic character is MnO<CrO<VO<TiO.
(iv) Mn_2O_7 is a covalent green oil.

(1) i, ii, iii, iv

(2) i, ii

(3) i, ii, iv

(4) iii, iv

9) Find the number of green colour compound(s) -
MnO_2, K_2MnO_4, AgI, $Cr_2(SO_4)_3$, $CuSO_4.5H_2O$, $CuCO_3.Cu(OH)_2$, $FeSO_4.7H_2O$

(1) 3

(2) 4

(3) 5

(4) 6

10) Select the **CORRECT** reactant for brown ring test of NO_3^- ion

(1) Aqueous solution of salt + ferrous sulphate solution + dil. H_2SO_4

(2) Aqueous solution of salt + ferric sulphate solution + conc. H_2SO_4

(3) Aqueous solution of salt + ferrous sulphate solution + conc. H_2SO_4

(4) Aqueous solution of salt + ferric sulphate solution + dil. H_2SO_4

11) Borax Bead test is carried out when the original mixture is coloured. It is done with the help of a clean platinum wire on which a small loop is made at the end. When borax is heated on platinum wire loop a transparent glass like bead is obtained. The hot bead is brought in contact with salt till it reacts with fused borax and colour is imparted to the bead. Bead colour is noted.

List-I		List-II	
(P)	Blue Green	(1)	Ni^{+2}
(Q)	Yellow	(2)	Cr^{3+}
(R)	Green	(3)	Fe^{3+}
(S)	Brown	(4)	Cu^{2+}

(1) P-(1), Q-(3), R-(2), S-(4)

(2) P-(4), Q-(3), R-(2), S-(1)

(3) P-(4), Q-(1), R-(3), S-(2)

(4) P-(1), Q-(4), R-(3), S-(2)

12)

How many elements having lesser size than Eu.
La, Yb, Pm, Ce, Nd, Gd, Ho, Tm

(1) 7

(2) 8

(3) 5

(4) 0

13)

For an element the successive ionisation energy values (in eV/atom), are given below.
14.534, 29.601, 47.448, 77.472, 97.888, 552.057, 667.029

Find the number of valence shell electrons in that element.

(1) 3

(2) 4

(3) 5

(4) 6

14) $BCl_3 + 3HOH \longrightarrow (X) + 3HCl$

In (X) total number of σ bond = a
Total number of lone pair of electron = b
Basicity of (X) is = c
if a + b + c = Z
Find the value of (Z)

(1) 7

(2) 9

(3) 10

(4) 13

15) Methanides are :

(1) Mg_2C_3, Be_2C, Al_4C_3 and CaC_2

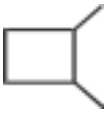
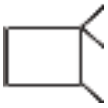
(2) Mg_2C_3, Be_2C and Al_4C_3

(3) Be_2C, Al_4C_3 and CaC_2

(4) Be_2C and Al_4C_3

16)

Match the column-I and column-II.

	Column-I		**Column-II**
	(Compound)		**Number of Stereoisomers**
(a)		(P)	2
(b)		(Q)	3
(c)		(R)	4
(d)		(S)	6

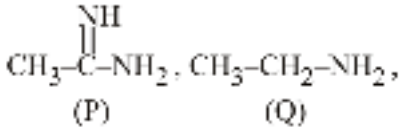
(1) (a) → (Q); (b) → (R); (c) → (P); (d) → (P)

(2) (a) → (R); (b) → (P); (c) → (Q); (d) → (P)

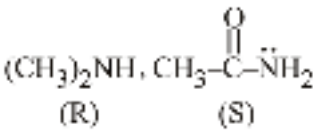
(3) (a) → (R); (b) → (Q); (c) → (P); (d) → (P)

(4) (a) → (R); (b) → (Q); (c) → (Q); (d) → (P)

17) Write basic strength order of following :-



(P) (Q)



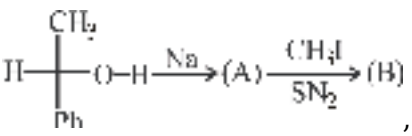
(R) (S)

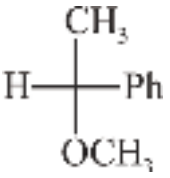
(1) P > S > R > Q

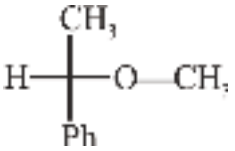
(2) R > S > Q > P

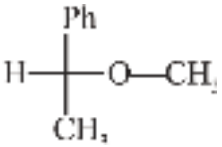
(3) P > R > Q > S

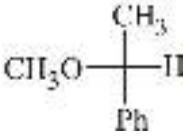
(4) Q > S > R > P

18)  , Product (B) is :-

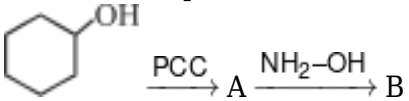
(1) 

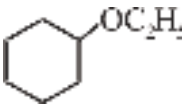
(2) 

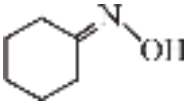
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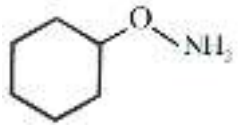
(4) 

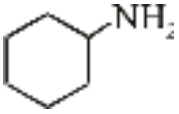
19) The final product (B) of the reaction sequence is :-



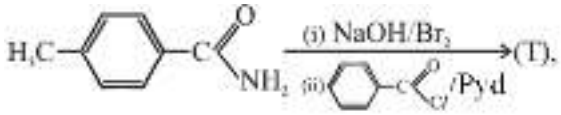
(1) 

(2) 

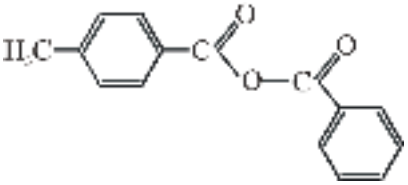
(3) 

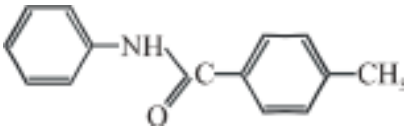
(4) 

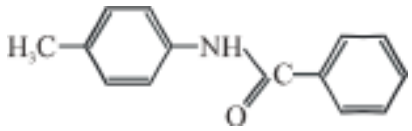
20) In the reaction,

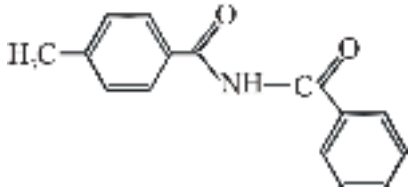


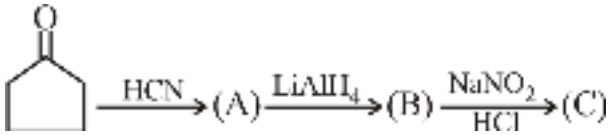
the structure of the product (T) is

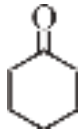
(1) 

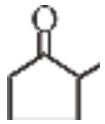
(2) 

(3) 

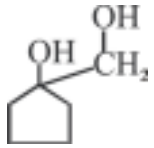
(4) 

21)  End product (C) in above reaction is :

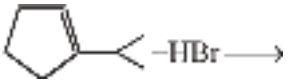
(1) 

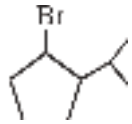
(2) 

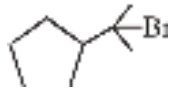
(3) 

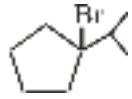
(4) 

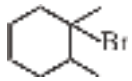
22) Which of the following major product is formed in the following reaction ?

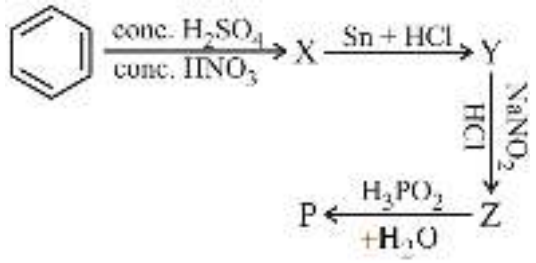


(1) 

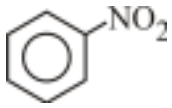
(2) 

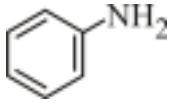
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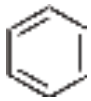
(4) 

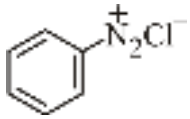
23) 

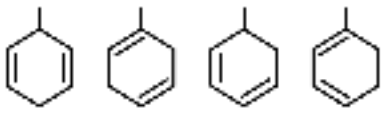
Compound P is :

(1) 

(2) 

(3) 

(4) 

24) Arrange the following in decreasing order of heat of hydrogenation. 

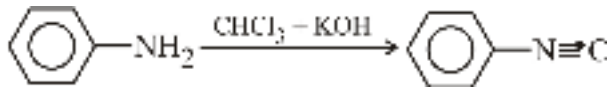
(1) P > Q > R > S

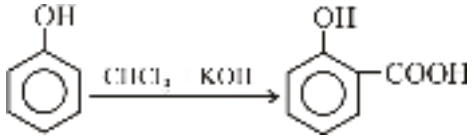
(2) S > R > Q > P

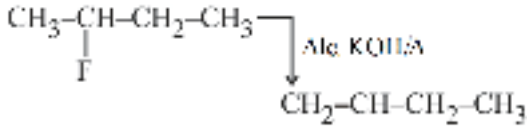
(3) Q > P > S > R

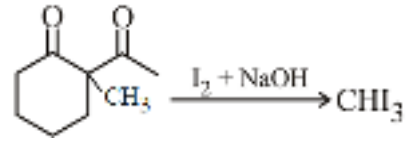
(4) R > Q > P > S

25) Which of the following reaction is incorrect :-

(1) 

(2) 

(3) 

(4) 

26) For the estimation of iodine by carius method 100 gram of an organic compound is taken in carius flask. If 47 gram of silver iodide (AgI) is obtained then precentage of iodine in given sample of organic compound is :-

(1) 2.54%

(2) 25.4%

(3) 54.2%

(4) 5.42%

27) **Assertion** - Glucose reacts with hydroxylamine to confirm the presence of carbonyl group in glucose.
Reason - Glucose forms addition product with $NaHSO_3$.

(1) Both assertion & reason are correct and reason is correct explanation of assertion.
(2) Assertion is correct, reason is not correct.
(3) Both assertion and reason are correct but reason is not correct explanation of assertion.
(4) Assertion is wrong, reason is correct.

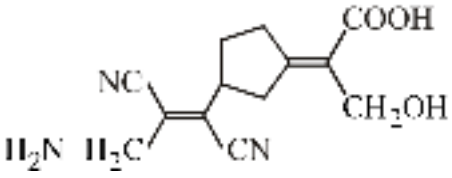
28) Which of the following statement(s) is incorrect :-

(1) Phenol does not react with $NaHCO_3$
(2) An aqueous solution of ethanol does not affect the litmus
(3) All the secondary alcohols gives haloform reaction
(4) The ease of the esterification of formic acid, acetic acid and propionic acid with ethanol follows the order - Formic acid > Acetic acid > propionic acid

29) Which of the following is correct IUPAC name.

(1) 4-Chloro-3-methyl cyclopentanol
(2) 1-Amino-3-bromo hexan-1-one
(3) 4-Chloro-3-methyl cyclohexane carboxylic acid
(4) 3-Bromo-1-methyl hexan-1-ol

30) Assign Configurations to the double bond in the given molecule :-



(1) E, Z
(2) Z, E
(3) E, E
(4) Z, Z

31)

Column-I		Column-II	
(A)	Number of carbon atoms in 90 g $C_6H_{12}O_6$	(P)	3
(B)	Number of g-atoms of oxygen in 1mol Fe_2O_3.	(Q)	$3N_A$
(C)	Mass of ozone (in g) having same number of molecules as there are in 1 mol of O_2.	(R)	48
(D)	Mass of oxygen (in g) in 0.25 mol of $Mg_2P_2O_7$.	(S)	32
		(T)	28

Select the correct match -

(1) (A) Q, (B) P, (C) R, (D) T

(2) (A) T, (B) R, (C) P, (D) Q

(3) (A) P, (B) Q, (C) R, (D) T

(4) (A) P, (B) Q, (C) T (D) R

32) Calculate pOH of aqueous solution at 25°C obtained by mixing equal volume of two aqueous solution having pH = 2 and pH = 4 respectively.[Take : log 5 = 0.7]

(1) 2.3

(2) 3.5

(3) 11.7

(4) 12.7

33) Which of the following solutions will have a pH greater than 7 at 298 K?
[Given : pK_a of CH_3COOH = pK_b of NH_4OH = 4.74]

(1) A solution containing equimolar amount of CH_3COOH and CH_3COONa.

(2) A solution obtained by mixing equal volumes of NaOH solution having molarity 0.1 M with HCl solution having molarity 0.2 M.

(3) A solution obtained by mixing equal concentration of NH_4OH & CH_3COOH in a volume ratio 1 : 1 respectively.

(4) A solution obtained on mixing equal concentrations of CH_3COOH and NaOH in a volume ratio of 1 : 2 respectively.

34) Sulphur undergo a phase transformation at 'T' K.

$S_8(\text{Rhombic}) \longrightarrow S_8(\text{Monoclinic})$

$\Delta_{rxn} H° = 3$ kJ/mol, $\Delta_{rxn} S° = 9\ JK^{-1}$

Then from given data :

(1) At 400 K Rhombic sulphur is more stable.

(2) At 300 K monoclinic is more stable.

(3) At 400 K monoclinic is more stable.

(4) both (2) and (3)

35) **Statement-1 :** If the normality of a $KMnO_4$ solution is 0.5N, then the normality of same solution may be 0.3N, in other condition.
Statement-2 : Equivalent weight of any substance may be different in different reaction condition.

(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

36)

Column-I		Column-II	
(A)	Radius of 2nd orbit of Li^{2+} ion	(P)	$\frac{K\pi e^2}{h}$
(B)	Velocity of electron in 4th orbit in He^+ ion	(Q)	$\frac{-13.6 \times 9}{4}$ eV
(C)	Total energy of electron in 2nd orbit of Li^{2+}	(R)	$\frac{h^2}{\pi^2 mKe^2 \times 3}$
(D)	Frequency of revolution of electron in 2nd orbit of Li^{2+} ion.	(S)	$\frac{-13.6 \times 4}{9}$eV
		(T)	$\frac{9\pi^2 mK^2 e^4}{2h^3}$

(1) (A) R (B) P (C) Q (D) T

(2) (A) T (B) Q (C) P (D) R

(3) (A) P (B) R (C) Q (D) T

(4) (A) R (B) P (C) T (D) Q

37) For the following heterogeneous phase equilibrium
$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + H_2O(g) + CO_2(g)$
Which of the following is not the favourable condition for dissociation of sodium bicarbonate($NaHCO_3$)?

(1) Addition of inert gas at constant pressure.

(2) Increase in temperature.

(3) Decrease in pressure at constant temperature.

(4) Addition of $CO_2(g)$ at constant volume.

38) The reaction : $A(g) + 2B(g) \rightarrow C(g) + D(g)$ is an elementary process. In an experiment, the initial partial pressures of A and B are $P_A = 0.80$ atm, $P_B = 0.80$ atm. When $P_C = 0.2$ atm, the rate of reaction relative to the initial rate is -

(1) 1/48

(2) 1/24

(3) 3/16

(4) 1/6

39) Identify the correct statement :

(1) Aquatic species are more comfortable in hot water than cold water due to higher solubility of $O_2(g)$ in hot water.

(2) A minimum boiling azeotrope will have higher vapour pressure than both of its components.

(3) If 0.1 M NaCl(aq) solution is diluted then its boiling point will remain same because number of particles of NaCl are constant.

(4) Osmotic pressure get changed if solvent is changed keeping same concentration.

40) If the conductivity of 0.001 M propionic acid is 3.83×10^{-5} S-cm^{-1} and limiting molar conductivities of HCl, KCl and potassium propionate are 426, 126, 83 S cm^2 mol^{-1} respectively then acid dissociation constant of propionic acid is -

(1) 10^{-6}

(2) 1.11×10^{-5}

(3) 1.11×10^{-4}

(4) 10^{-3}

41) Which of the following statement(s) is /are **incorrect**?

(1) Metal sol can be prepared by Bredig's arc method

(2) Lyophilic colloids are stable due to extensive solvation.

(3) Peptization is the process of converting a precipitate into colloidal solution.

(4) Colloid of starch is an example of Lyophobic sol.

42) Standard enthalpy of formation of $CH_3OH(\square)$ is :

Given : $H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\square)$

$\Delta_f H° = -285$ kJ/mol

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)$

$\Delta_C H° = -393$ kJ/mol

$CH_3OH(\square) + \frac{3}{2} O_2(g) \rightarrow CO_2(g) + 2H_2O$

$\Delta H° = -725$ kJ/mol

(1) –138 kJ/mol

(2) 47 kJ/mol

(3) –238 kJ/mol

(4) – 47 kJ/mol

43) For the first order reaction, $A(g) \rightarrow 2B(g) + C(s)$, $t_{1/2} = 24$ minute. The reaction is carried out taking certain mass of A enclosed in a vessel having constant volume in which it exerts a pressure of 400 mm Hg. The pressure of the reaction mixture after expiry of 48 minute will be

(1) 700 mm Hg

(2) 600 mm Hg

(3) 800 mm Hg

(4) 1000 mm Hg

44) Identify the option which is correct :

(1) for an adiabatic process, entropy of system must remain constant always.

(2) for chemical reaction to be spontaneous Gibbs free energy should increase.

(3) for any substance heat required for melting will always be equal to heat required for vaporisation.

(4) $\Delta S_f°$ of $H_2O_{(\square)}$ is < 0

45) A mixture of $NaHC_2O_4$ and $H_2C_2O_4$ requires 50 mL, 0.1 M $KMnO_4$ (aq.) solution during titration in acidic medium. The same mixture of $NaHC_2O_4$ and $H_2C_2O_4$ requires 50 mL, 0.4 M NaOH (aq.) solution for the complete neutralisation. Calculate the mass of $H_2C_2O_4$ in the initial mixture.

(1) 1.250 g

(2) 0.900 g

(3) 0.450 g

(4) 0.675 g

BIOLOGY

1) In Urey and Miller experiment which of the following combination was used ?

(1) H_2O (vapour), CH_4, O_2, NH_3

(2) H_2O (vapour), CH_4, NH_3, CO

(3) CH_4, NH_3, H_2, CO

(4) CH_4, NH_3, H_2, H_2O (vapour)

2) **Assertion (A) :** Population sizes in reality are stable. Except seasonal variation
Reason (R) : In nature there had been competition for resources which are limited.
Choose the correct answer from the given options.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) Both (A) and (R) are incorrect.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

3) How many of the following are characters of withdrawal syndrome ?
Shakiness, High fever, Runny nose, Sweating, Mental retardation, Anxiety, Nausea.

(1) Three

(2) Four

(3) Five

(4) Six

4) AIDS is caused by HIV. among the following, which one is not a mode of transmission of HIV ?

(1) Transfusion of contaminated blood

(2) Sharing the infected needles

(3) Shaking hands with infected person

(4) Sexual contact with infected person

5) **Statement-I :** Filarial worms cause a slowly developing chronic inflammation in blood vessels of the lower limbs.
Statements-II : Maintenance of personal and public hygiene is very important for prevention and control of non- infectious diseases.

(1) Only statement-I is correct

(2) Only statement-II is correct

(3) Both statements are correct

(4) Both statements are incorrect

6) Read the following statements :
(a) Acquired immunity is pathogen specific.
(b) Acquired immunity is characterised by memory.
(c) The primary and secondary immune responses are carried out with the help of only one special type of lymphocyte present in our body i.e., B-lymphocytes.
(d) T-cells themselves do not secrete antibodies but help the B cells to produce them
(e) Humoral immune response is mediated by B lymphocytes
Select the option which includes all the correct statements only

(1) a,b,d and e

(2) b,c,d and e

(3) a,b,c and d

(4) a,b,c,d and e

7) Match Column - I with Column - II.

Column - I		**Column - II**	
(a)	LNG-20	(i)	Copper releasing IUD
(b)	Lippes loop	(ii)	Hormone releasing IUD
(c)	Multiload 375	(iii)	Barrier method
(d)	Condoms	(iv)	Non medicated IUD

Choose the correct answer from the options given below :

(1) (a) – (iv), (b) – (i), (c) – (iii), (d) – (iii)

(2) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)

(3) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)

(4) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)

8) The type of placenta found in human is formed by interdigitation of :

(1) Umbillical cord and chorionic villi

(2) Myometrium and Endometrial tissue

(3) Chorionic villi and uterine tissue

(4) Epithelium of amnion and chorion

9) Which of the following statement is incorrect ?

(1) Presence or absence of hymen is a reliable indicator of virgnity or sexual experience

(2) Several mammary ducts join to form a wider mammary ampulla.

(3) During pregnancy, all events of the menstrual cycle stop and there is not menstruation.

(4) In the absence of fertilization, the corpus luteum starts degenerating.

10) Which of the following event takes place after entry of sperm into the ovum ?

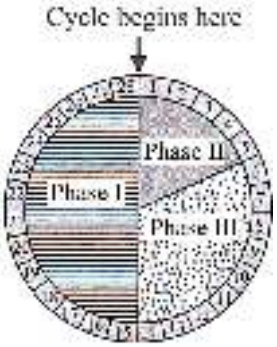
(1) Completion of meiotic division of secondary oocyte

(2) Formation of second polar body

(3) Formation of a haploid ootid

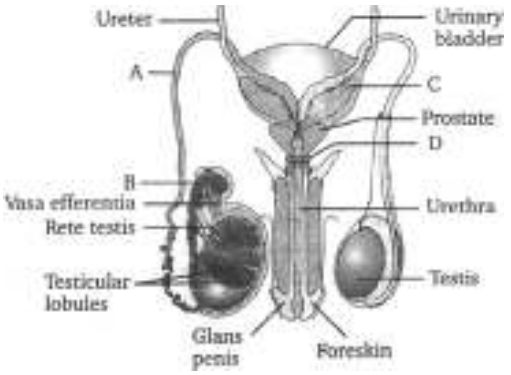
(4) All of these

11) Refer the given schematic representation of menstrual cycle showing three phases I, II and III.



Select the option that correctly matches these phases with the following events.
(i) FSH secreted by the anterior pituitary stimulates the ovarian follicle to secrete oestrogen which stimulates the proliferation of endometrium.
(ii) LH secreted by the anterior pituitary stimulates the development of corpus luteum which secretes progesterone
(iii) Reduced production of LH from anterior pituitary causes degeneration of corpus luteum, thereby menustral flow starts.

(1)	Phase - I (i)	Phase - II (ii)	Phase - III (iii)
(2)	Phase - I (iii)	Phase - II (i)	Phase - III (ii)
(3)	Phase - I (ii)	Phase - II (iii)	Phase - III (i)
(4)	Phase - I (iii)	Phase - II (ii)	Phase - III (i)



12) Which of the following is correct for A, B, C and D

(1) A-Carries sperm,B-Stores sperm permanently, C-Secretes a fluid that lubricates ureter, D-Secretes a fluid that ultimately becomes semen.

(2) A-Carries sertoli cells,B-Stores sperm, C-Carries urine, D-Stores sperm.

(3) A-Carries sperm, B-Stores sperm, C-Secretes a fluid that ultimately becomes semen, D-Secretes a fluid that lubricates penis.

(4) A-Carries sperm, B-Stores sperm permanently, C-Secretes a fluid that lubricates distal urethra, D-Secretes a fluid that ultimately becomes semen

13) Which type of movement is present in trachea ?

(1) Amoeboid movement

(2) Flagellar movement

(3) Ciliary movement

(4) Muscular movement

14) Choose the correct statements :
(A) Each rib has two articulation surface on its ventral end and is hence called bicephalic.
(B) In human number of ear ossicles is three.
(C) Dorsally ribs are attached to the thoracic vertebrae.
(D) A cup shaped bone called patella cover the knee ventrally.

(1) A, B & C

(2) A, B, C & D

(3) B & C

(4) C & D

15) Match the column - I with column - II and select the correct option.

Column - I		**Column - II**	
A.	Somatostatin	I.	Pancreas
B.	Acromegaly	II.	Posterior pituitary
C.	Diabetes insipidus	III.	Hypothalamus
D.	Diabetes mellitus	IV.	Anterior pituitary

(1) A – III, B – II, C – IV, D – I

(2) A – II, B – IV, C – I, D – III

(3) A – I, B – IV, C – II, D – III

(4) A – III, B – IV, C – II, D – I

16) Read the following statements are :
a. In adult women, Hypothyroidism may cause menstrual cycle to become irregular.
b. Oxytocin acts on the smooth muscles of our body and stimulates their contraction.
c. MSH acts on the melanocytes and regulates pigmentation of the skin.
d. The pineal gland is located on the ventral side of forebrain.
e. Parathyroid hormone is a hypocalcemic hormone.
Choose the option with correct statements only :

(1) a, b and d

(2) b, c and e

(3) a, b, c and e

(4) a, b and c

17) Which one of the following hormone secreted by ovary and acts on uterus?

(1) FSH

(2) LH

(3) Progesterone

(4) Cortisol

18) Exophthalmic goiter is characterized by _____ of the thyroid gland, protrusion of eye balls, increased BMR & weight _____.

(1) enlargement, loss

(2) gain, atrophy

(3) enlargement, gain

(4) atrophy, loss

19) **Assertion :** At resting state, positive charge is more outer surface and negative charge more inner surface of the axon.
Reason : Charge difference around axolemma arises due to active transport only in resting state.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

20) When a Neuron is not conducting any impulse i.e, resting, the axonal membrane comparatively _____ to potassium ions (K^+) and nearly _____ to sodium (Na^+) ions.

(1) Less permeable, impermeable respectively

(2) More permeable, impermeable respectively

(3) Less permeable, more impermeable respectively

(4) Impermeable, impermeable respectively

21) The blood vessels supplying blood into Bowman's capsule is –

(1) Afferent arteriole

(2) Efferent arteriole

(3) Renal vein

(4) Renal portal vein

22) **Statement-I :-** Malfunctioning of kidneys can lead to accumulation of ammonia in blood, a condition called uremia, which is highly harmful and may lead to kidney failure.
Statement-II :- In such patients, urea can be removed by a process called hemodialysis.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I is correct but II incorrect

(4) Statement-I is incorrect but II correct.

23) **Statement-I :** The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.
Statement-II : The overall increase in the thoracic volume causes a similar increase in pulmonary volume.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement I is correct but II incorrect

(4) Statement I is incorrect but II correct.

24) How much Partial pressure of O_2 is in Deoxygenated blood ?

(1) 104

(2) 159

(3) 40

(4) 95

25) The QRS complex represent the _____ which initiates the ventricular contraction.

(1) Depolarisation of Atria

(2) Depolarisation of Ventricles

(3) Repolarisation of Ventricles

(4) Repolarisation of Atria

26) Read following terminology of column-I & II and identify correct match :-

	Column-I		**Column-II**
(A)	Diastolic pressure	(i)	Electrocardiogram
(B)	Heart rate	(ii)	70 ml/min by both ventricle
(C)	Thrombo plastin	(iii)	72 beat per min
(D)	End of Twave	(iv)	Start of ventricular diastole
		(v)	Helps in clotting
		(vi)	80 mm of Hg
		(vii)	Prevent clotting.

(1) A-(i), B-(iv), C-(ii), D-(vi)

(2) A-(i), B-(ii), C-(iv), D-(v)

(3) A-(ii), B-(iii), C-(iv), D-(v)

(4) A-(vi), B-(iii), C-(v), D-(iv)

27) In frog, mature female can lay how many ova at a time ?

(1) 2500-3000
(2) 1500-2500
(3) 1500-3000
(4) 1000-3000

28) Epithelial tissue faces either a _____ or the _____ and thus provides a covering or a lining for some part of the body.

(1) External fluid, body environment.
(2) Skin, organs
(3) Body, environment.
(4) Body fluid, external environment

29) Which of the following is correct statement about cockroach ?

(1) A pair of unequal sized spermatheca are associated with genital chamber.
(2) Spermatheca are part of male reproductive system and they are sperm production site.
(3) Sperms are transferred from spermatheca to spermatophore.
(4) A pair of unequal sized spermatheca are present in 7th-8th abdominal segments of female cockroach.

30) In which phylum segmentation is Absent ?

(1) Chordata
(2) Aschelminthes
(3) Annelida
(4) Arthropoda

31) Phylum Aschelminthes exhibit -

(1) Tissue level organisation, triploblastic, pseudocoelom
(2) Organ system level organisation, Triploblastic
(3) pseudocoelom, diploblastic, cell aggregate
(4) Tissue level organisation, Acoelomate

32) Match the following :-

(A)	Echinodermata	(i)	Visceral Hump
(B)	Mollusca	(ii)	Proboscis Gland
(C)	Arthropoda	(iii)	Malpighian Tubules
(D)	Hemichordata	(iv)	Nephridia
(E)	Annelida	(v)	Ambulacral (Water vascular system)

(1) (A) - (i), (B) - (v), (C) - (iii), (D) - (iv), (E) - (ii)

(2) (A) - (v), (B) - (iv), (C) - (iii), (D) - (i), (E) - (ii)

(3) (A) - (iii), (B) - (ii), (C) - (v), (D) - (iv), (E) - (i)

(4) (A) - (v), (B) - (i), (C) - (iii), (D) - (ii), (E) - (iv)

33) You are observing an animal closely and you find some characters such as -
* Endoskeleton made up of cartilage
* Mouth is located ventrally
* The skin is tough, containing placoid scales
* Notochord persist throughout life
* Air bladder is absent

After observing the above features in which group will you classify that animal :-

(1) Chondrichthyes

(2) Osteichthyes

(3) Cyclostomata

(4) Cephalochordata

34) Read the following statements :-
(a) Sponges reproduce sexually by fragmentation and asexually by the formation of gametes.
(b) In cnidarians the digestion is extracellular and intracellular both.
(c) In ctenophores reproduction takes place only by asexual means.
(d) In platyhelminthes fertilisation is internal and development is indirect.
(e) In *Nereis*, parapodia help in swimming.
Choose the correct options only :-

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (b), (d) and (e) only

(4) (a), (b), (c) and (d)

35) A cell organelle that is present in animal cells but not present in plants cells is:-

(1) Chloroplast

(2) Mitochondria

(3) Centrosome

(4) Golgibody

36) **Assertion (A) :-** Mitochondria and chloroplasts are not considered part of the endomembrane system despite being membrane-bound organelles.
Reason (R) : The endomembrane system includes only those organelles which are related with energy metabolism & whose function are not co-ordinated with ER
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both A and R are true and R explains A

(2) Both A and R are true but R does not explains A

(3) A is true but R is false

(4) A is false but R is true

37) From the statement given below choose the correct option:-
A - RER is frequently observed in the cells actively involved in steroidal hormones synthesis.
B - The golgi apparatus principally performs the function of Packaging materials.
C - Lysosomes are membrane bound vesicular structures formed by the process of Packaging in the golgi apparatus.
D - Golgi apparatus is the important site of formation of glycoproteins and glycolipids.
E - In Amoeba the food vacuoles in important for osmoregulation and excretion.

(1) A, B, C are true

(2) A, B, D are true

(3) A, B, E are true

(4) B, C, D are true

38) Meiosis involves :-

(1) Two nuclear divisions and one chromosome division.

(2) One nuclear division and one chromosome division.

(3) One nuclear division and two chromosome divisions.

(4) Two nuclear division and three chromosome divisions.

39) Which of the following is the event in the G_1-phase?

(1) DNA synthesis

(2) Cell division

(3) Cell growth and preparation for DNA Replication

(4) Chromosome duplication

40) Read the given passage and answer the following questions :-
"K" is a uniquely long and complex stage of cell division where homologous chromosomes pair up and exchange genetic material. Based on chromosomal behaviour, "K" is further divided into five sub-stages.
Select the correct sequence of the events occurring during "K" :-
W-Exchange of genetic material between non-sister chromatids
X-Dissolution of synaptonemal complex.
Y-Formation of synaptonemal complex and synapsis.
Z-Chromosomes fully condensed and initiation of meiotic spindle assembly.

(1) Y → W → X → Z

(2) W → Y → Z → X

(3) Z → Y → W → X

(4) X → Z → Y → W

41) Antibiotic resistance gene present with Bam HI site of a *E.Coli* cloning vector is :-

(1) Ampicillin resistance gene

(2) Tetracycline resistance gene

(3) Chloramphenical resistance gene

(4) Kanamycin resistance gene

42) Match the columns and choose correct option ?

	Column I		Column II
a	Recombinant technologies	i	Large volumes (100-1000 litres) of culture can be processed
b	Continuous culture system	ii	The ultimate aim is to produce a desirable protein
c	Bioreactors	iii	Used to produces a larger biomass leading to higher yields of desired products

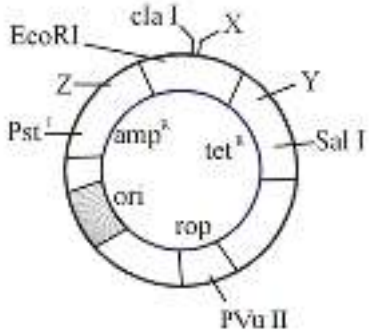
(1) a-i, b-ii, c-iii

(2) a-ii, b-iii, c-i

(3) a-iii, b-i, c-ii

(4) a-ii, b-i, c-iii

43) Identify the restriction enzyme used to cut the site X, Y and Z respectively.



(1) Pst II, Bam HI, ECOR II

(2) Hind III, Bam HI, Pvu I

(3) Hind II, Bam HII, Pst II

(4) Hind II, ECOR II, Pvu I

44) Select the correct option –
(i) Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs.
(ii) Mature insulin consists of two short polypeptide chains.
(iii) In mammals, including humans, insulin is synthesized as pro-hormone.
(iv) Insulin from an animal source caused some pateint, to develop allergy,
How many statements are correct.

(1) Two

(2) Three

(3) Four

(4) One

45)

Match the column I with column II :

	Column I		**Column II**
A	Baculovirus	i	Phosphorus nutrition
B	*Mycorrhiza*	ii	*Nucleopolyhedrovirus*
C	*Trichoderma*	iii	Free living fungi
D	*Cyanobacteria*	iv	Biofertiliser for paddy field

(1) A-ii, B-i, C-iii, D-iv

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-i, D-ii

(4) A-i, B-ii, C-iv, D-iii

46) Which of the following statements are true ?
A. An organism represent/occupies a place or position in the system of classification.
B. One specific epithet may be present in different genus representing different species.
C. Human belongs to Homo species.
D. Animals, mammals and dogs represent the taxa at same level.
options:

(1) A, B and C

(2) A, B, C and D

(3) A, B and D

(4) A and B

47) Which one statement is wrong with regarding to dinoflagellates ?

(1) They are mainly marine

(2) Found on surface of water

(3) Nutrition mainly holozoic

(4) They appear yellow, green brown in colour.

48) Match list-I with list-II

List-I		**List-II**	
(A)	*Neurospora*	(i)	LSD source
(B)	*Claviceps*	(ii)	Edible
(C)	*Penicillium*	(iii)	Biochemical and genetic work
(D)	Truffles	(iv)	Antibiotic

	A	**B**	**C**	**D**

(1)	i	ii	iii	iv
(2)	iii	i	iv	ii
(3)	ii	iii	i	iv
(4)	iv	iii	ii	i

(1) 1

(2) 2

(3) 3

(4) 4

49) Match the organisms with their habitats :

	Column-I		**Column-II**
(A)	Halophiles	(I)	Marshy area
(B)	Thermoacidophiles	(II)	Fresh water
(C)	Methanogen	(III)	Hot spring
(D)	Chrysophytes	(IV)	Marine environment
		(V)	Extreme salty area
		(VI)	Decaying twigs and leaves
		(VII)	Moist soil

(1) A-I,II, B-III,V, C-V, D-VI,VII

(2) A-V, B-I, C-III, D-II,VI

(3) A-V, B-III, C-I, D-II,IV

(4) A-III, B-I, C-V, D-II,VII

50) Which of the following statements is/are correct regarding life cycle of pteridophyte ?
(A) The sporophyte bear the sporangia that are subtended by leaf like structure called sporophyll.
(B) The sporangia produce spore by mitosis in spore mother cell.
(C) The spore germinate to give rise to conspicuous small but multicellular free living mostly photosynthetic thalloid gametophyte called prothallus.
(D) Gametophyte require cool damp shady places to grow.

(1) Option A,B,C

(2) Option B,C,D

(3) Option A,C

(4) Option A,D

51) Match List-I with List-II

List-I		**List-II**	
(a)	Multicellular branched Rhizoids	(i)	*Porphyra*

(b)	Flagellated, Isogamous Reproduction	(ii)	*Spirogyra*
(c)	Non flagellated, Isogamous Reproduction	(iii)	*Ulothrix*
(d)	Floridean starch	(iv)	*Sphagnum*

(1) (a)-(iv) (b)-(iii) (c)-(ii) (d)-(i)

(2) (a)-(iii) (b)-(iv) (c)-(ii) (d)-(i)

(3) (a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)

(4) (a)-(iv) (b)-(ii) (c)-(iii) (d)-(i)

52) Find out the correct match from the following.

	Hypogynous flower	**Perigynous flower**	**Epigynous flower**
1	Guava, Cucumber, Ray florets of Sunflower	Plum, Rose, Peach	Mustard, Brinjal, Chinarose
2	Plum, Rose, Brinjal	Cucumber, Peach	Guava, Ray florets of Sunflower
3	Mustard, Guava	Brinjal, Ray florets of Sunflower	Cucumber, Chinarose
4	Mustard, Brinjal, Chinarose	Plum, Rose, Peach	Guava, Cucumber, Ray florets of Sunflower

(1) 1

(2) 2

(3) 3

(4) 4

53) Edible part of mango and coconut is respectively :

(1) Mesocarp and endocarp

(2) Mesocarp and endosperm

(3) Epicarp and endosperm

(4) Endocarp and endosperm

54) Among china rose, tomato, lemon, *Argemone*, mustard, *Dianthus*, Primrose, sunflower, pea and marigold, In how many plants, the placenta develops at the base of ovary and a single ovule is attached to it ?

(1) Three

(2) Two

(3) Four

(4) Five

55) The epidermal cells of monocot roots are :-

(1) Chlorenchymatous

(2) Parenchymatous
(3) Sclerenchymatous
(4) Collenchymatous

56) Hypodermis of sunflower stem is :

(1) Sclerenchymatous
(2) Collenchymatous
(3) Parenchymatous
(4) Chlorenchymatous

57) Large number of small seeds are present in fruit of :

(1) *Orobanche* and *Striga*
(2) Watermelon and Mango
(3) *Ficus* and *Date palm*
(4) Orchids and Mango

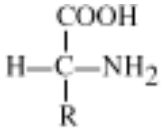
58) Which of the following is correctly matched ?

(1) Epidermis of anther: 2n
(2) Generative cell of pollen grain: n
(3) Middle layer: 2n
(4) All of the above

59) Arrange the following structures from largest to smallest in two generations of the same angiosperm.
(1) Ovary
(2) Ovule
(3) Egg
(4) Carpel
(5) Embryo sac

(1) (4), (2), (1), (5), (3)
(2) (5), (4), (3), (1), (2)
(3) (5), (1), (4), (2), (3)
(4) (4), (1), (2), (5), (3)

60) Which of the following amino acid is formed when the methyl group is substituted from the R group in the given formula ?



(1) Tryptophan
(2) Valine
(3) Glycine

(4) Alanine

61) Match the **Column I** with **Column II** & choose the correct option :-

	Column-I		**Column-II**
(A)	Collagen	(i)	Most abundant protein in world
(B)	Trypsin	(ii)	Most abundant protein in animal world
(C)	Insulin	(iii)	Proteolytic enzyme
(D)	RuBisCO	(iv)	Hormone

(1) A-iii, B-ii, C-iv, D-i

(2) A-i, B-iii, C-iv, D-ii

(3) A-ii, B-iii, C-iv, D-i

(4) A-iii, B-ii, C-i, D-iv

62) Read the following statements and select the set of **incorrect** statements :
(A) Nucleotides are the building blocks of nucleic acids.
(B) Protein, Nucleic acids and polysaccharides are considered biomacromolecules in living system.
(C) Lipids are classified as biomacromolecules because they are polymers.
(D) Polysaccharides are components of cell wall in plants, fungi and also exoskeleton of arthropods.
(E) The most abundant chemical in living organisms is protein.

(1) B and E only

(2) C and E only

(3) A, B and D only

(4) A, D and E only

63) Which of the following is an **incorrect** statement ?

(1) The protein portion of the enzyme is called cofactor.

(2) Prosthetic groups are tightly bound to the apoenzymes

(3) Co-enzymes are organic compounds

(4) Many enzymes require metal ions for their activity which form coordination bonds with side chains at the active site.

64) CO_2 saturation point in tomato and bell pepper plant respectively, is -

(1) at 360 μ/lit and beyond 450 μ/lit

(2) at 360 μ/lit and at 360 μ/lit

(3) beyond 450 μ/lit and at 360 μ/lit

(4) beyond 450 μ/lit and beyond 450 μ/lit

65) In C_4 plants, fixation of each molecule of CO_2 in Calvin cycle requires :

(1) 5 ATP and 2 NADPH

(2) 3 ATP and 2 NADPH

(3) 6 ATP and 2 NADPH

(4) 18 ATP and 12 NADPH

66) What is the net ATP production when two molecule of glucose are fermented to lactic acid ?

(1) Two

(2) Eight

(3) Ten

(4) Four

67) In tricarboxylic acid cycle, formation of isocitrate is followed by two successive steps of decarboxylation. The second decarboxylation results in the formation of :-

(1) Succinyl CoA

(2) Succinic acid

(3) Oxaloacetic acid

(4) α-Ketoglutaric acid

68) There are some steps in the respiratory process to release and utilise the energy stored in $NADH+H^+$ and $FADH_2$. This can achieved if $NADH+ H^+$ and $FADH_2$ act as electron donors and their electron are passed on to final acceptor in the arobic respiration process. The metabolic pathway through which all these process can occur, is present :

(1) in the matrix of the mitochondria

(2) in the cytoplasm of the cell

(3) in the inner mitochondrial membrane

(4) in the outer mitochondrial membrane

69) Fruit and leaf drop at early stages can be prevented by the application of :

(1) Ethylene

(2) Auxins

(3) Gibberellic acid

(4) Cytokinins

70) Which of the following plant growth regulator helps to produce new leaves and chloroplasts in leaves ?

(1) Auxin

(2) Gibberellin

(3) Ethylene

(4) Cytokinin

71) Which of the following is **incorrect** regarding the discovery of plant growth regulators (PGRs) ?

(1) Natural cytokinins were discovered from coconut milk also.

(2) Ethylene was identified as a volatile substance influencing fruit ripening.

(3) Gibberellins were discovered during the study of the 'bakanae' disease in wheat.

(4) Auxin was isolated by F.W. went from tips of coleoptiles of oat seedlings.

72) What distinguishes commensalism from parasitism ?

(1) Parasitism benefits both species

(2) In parasitism, interaction is detrimental to parasite

(3) Commensalism harms the host

(4) Parasitism harms the hosts, commenslism does not

73) **Assertion (A) :** Resource (food and space) availability is obviously essential for the unimpeded growth of a population.
Reason (R) : Ideally, when resources in the habitat are unlimited, each species has the ability of realise fully its innate potential to grow in number.

(1) Both A & R are correct and R is correct the explanation of A

(2) Both A & R are correct but R is not the correct explanation of A

(3) Assertion is true but Reason is incorrect

(4) Both Assertion and Reason are correct

74) _____ is the process by which inorganic nutrients are released from organic matter during decomposition.

(1) Humification

(2) Stratification

(3) Leaching

(4) Mineralisation

75) Common herbivores in aquatic ecosystem, are :-

(1) Insects

(2) Birds

(3) Molluscs

(4) Mammals

76) Select the **incorrect** statement from following :-

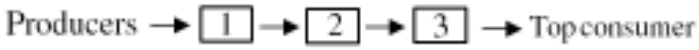
(1) Limited sunlight penetration is one of the reasons for low productivity of oceans.

(2) Primary productivity also depends on availability of nutrients.

(3) Oceans contribute 70 percent of the annual net primary productivity of the whole biosphere.

(4) Nutrient nitrogen is also a limiting factor for low productivity of oceans

77) Given below is the food chain Identify **correct** statements according to this food chain :



I. Zooplanktons, Mollusc, cattles belongs to [1] level
II. Man, Lion, Peacock can be top consumers
III. Fishes can be placed at both level [2] and [3]
IV. Sparrow can belongs to only level [3]
V. Wolf, fox, Jackal may belongs to level [2].

(1) I, II, IV only
(2) All of the above
(3) II, III, IV only
(4) I, II, III, IV only

78) Which of the following option correctly represent decreasing biodiversity order of vertebrates ?

(1) Birds > Amphibians > Fishes > Reptiles
(2) Fishes > Mammals > Birds > Amphibians
(3) Mammals > Birds > Fishes > Reptiles
(4) Fishes > Birds > Reptiles > Amphibians

79) Given below are two statements :
Statement-I : The *Clarias gariepinus* introduced into lake victoria in east africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
Statement-II : The recent illegal introduction of the african catfish nile perch for aquaculture purposes is posing a threat to the indigenous catfishes in Indian rivers.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

80) Match the columns with respect to Amazonian rain forest,

Column-I		**Column-II**	
A.	Plants	1.	125000
B.	Fishes	2.	427
C.	Birds	3.	1300
D.	Mammals	4.	378
E.	Reptiles	5.	40000
F.	Invertebrates	6.	3000

	A	**B**	**C**	**D**	**E**	**F**

1.	5	6	3	2	4	1
2.	5	6	4	3	2	1
3.	3	6	2	4	5	1
4.	6	5	4	2	1	3

(1) 1

(2) 2

(3) 3

(4) 4

81) Inheritance of skin colour in Human is an example of :

(1) Chromosomal aberration

(2) Point mutation

(3) Co - dominance

(4) Polygenic inheritance

82) Match the column I & II.

Column-I		**Column-II**	
(1)	Sex limited characters	(A)	X–linked gene
(2)	Sex influenced character	(B)	Y–linked gene
(3)	TDF/SRY gene	(C)	Pattern Bald-ness
(4)	Colour blindness	(D)	Beard moustache

(1) 1 – D, 2 – C, 3 – B, 4 – A

(2) 1 – A, 2 – B, 3 – C, 4 – D

(3) 1 – D, 2 – B, 3 – C, 4 – A

(4) 1 – B, 2 – C, 3 – A, 4 – D

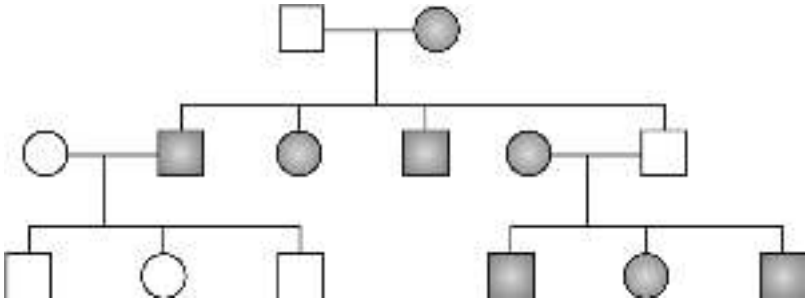
83) **Statement I :** Morgen coined term linkage.
Statement II : Tightly liked gene showed very low recombination.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

84) Below given pedigree belongs to : 

(1) X - linked dominant disease
(2) X - linked recessive disease
(3) Autosomal recessive disease
(4) Maternal inheritance

85) In which of the following life processes RNA is used as a catalyst ?

(1) translation
(2) splicing
(3) cleaving of RNA
(4) All of the above

86) In capping process :-

(1) An unusual nucleotide 7 methyl guanosine triphosphate is added to the 5' end of hn-RNA
(2) Adenylate residues are added at 3' end of hnRNA
(3) Removal of exons
(4) Joining of introns

87) A phenotypic ratio of 3 : 1 in the offspring of a mating of two organism heterozygous for a single trait is expected where.

(1) The allele segregate during meiosis.
(2) Each allele contains two mutation.
(3) The alleles are identical.
(4) The alleles are incompletely dominance

88) How many statements are incorrect –
(i) All three RNAs are needed to synthesize a protein in a cell.
(ii) RNA polymerase binds to terminator and initiates transcription.
(iii) In prokaryotes three types of RNA polymerase enzyme present.
(iv) The mRNA provides the template for protein synthesis.

(1) Two
(2) Three
(3) One
(4) Four

89) **Assertion (A):** Association with initiation factor (σ) and termination factor (ρ) alter the

specificity of the RNA polymerase to either initiate or terminate.
Reason (R): Severo Ochoa enzyme (polynucleotide phosphorylase) was also helpful in polymerising RNA with defined sequence in a template independent manner.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

90) The following are the few examples of eukaryotic RNAs. How many of them is/are transcribed by RNA polymerase-II ?
5s rRNA, 5.8s rRNA, 18s rRNA, 28s rRNA, hnRNA, tRNA, SnRNA :

(1) Three

(2) One

(3) Four

(4) Two

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	4	1	3	2	2	1	2	2	1	2	1	3	2	4	1	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	4	2	4	2	1	1	4	3	2	2	1	3	4	4	2	1	4	3
Q.	41	42	43	44	45															
A.	2	2	1	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	1	2	3	3	1	3	2	3	2	2	3	4	4	3	3	2	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	1	2	2	2	3	3	3	1	3	4	3	1	1	4	3	2	2
Q.	86	87	88	89	90															
A.	4	3	1	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	2	3	4	1	3	3	1	4	3	3	3	4	4	4	3	1	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	4	1	3	2	4	1	4	1	2	2	4	1	3	3	3	4	1	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	2	3	1	4	3	2	3	4	1	4	2	2	2	2	1	4	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	1	4	2	4	1	3	2	4	3	4	1	4	3	3	4	4	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	1	4	4	4	1	1	1	2	2										

SOLUTIONS

PHYSICS

1) By using dimensional analysis
$ML^2T^{-2} = M^x(T^{-1})^y(L)^z = M^xL^zT^{-y}$
$\Rightarrow x = 1, z = 2, y = 2$
$\Rightarrow 2x + y + z = 2 \times 1 + 2 + 2 = 6$

2) $F_1 + F_2 = 10$
$F_1 - F_2 = 6 \Rightarrow F_1 = 8, F_2 = 2$
New $F_1^1 = 8 + 3 = 11$
$F_2^1 = 2 + 3 = 5$
$R = \sqrt{11^2 + 5^2 + 2 \times 11 \times 5\cos 90^\circ}$
$= \sqrt{121 + 25} = \sqrt{146}$

3)

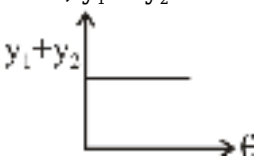
$$\text{Least Count} = \frac{\text{Pitch}}{\text{No. of divisions}} = \frac{0.5}{50} = 0.01\text{mm}$$

Zero Error = No. of divisions × Least Count

Zero Error = –[50–45] × L.C.
= –5 × 0.01 = –0.05 mm

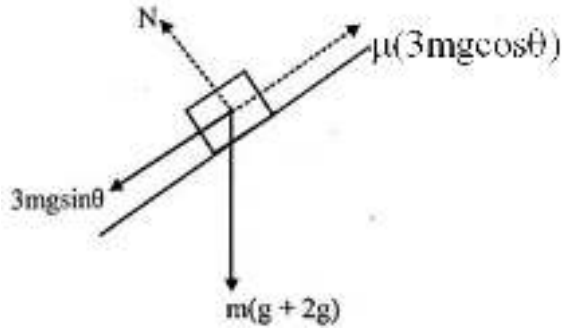
4) $\vec{v} = t\hat{i} + t^2\hat{j} = |\vec{v}| = \sqrt{t^2 + t^4} = t\sqrt{1 + t^2}$
Rate of change of speed
$$= \frac{d|\vec{v}|}{dt} = \sqrt{1 + t^2} \times 1 + \frac{1}{2}\left(1 + t^2\right)^{-\frac{1}{2}} \times 2t^2$$
$$= \sqrt{1 + t^2} + \frac{t^2}{\sqrt{1 + t^2}} = \frac{1 + 2t^2}{\sqrt{1 + t^2}}$$

5) $y_1 = \dfrac{u^2\sin^2\theta}{2g}$
$y_2 = \dfrac{u^2\cos^2\theta}{2g}$
Now, $y_1 + y_2 = \dfrac{u^2}{2g}$ = constant



6)

Let μ be the minimum coefficient of friction.
At equilibrium, mass does not move so,
3 mg sin θ = μ(3mgcosθ)
$\therefore \mu_{min} = \tan\theta$



7)

It happens due to inertia of rotation.

8) $W_{Net} = \Delta KE$

$$W_{Net} = \frac{1}{2}mv_F^2 - \frac{1}{2}mv_i^2$$
$$= \frac{1}{2}m\left[\alpha^2 x_0^3\right] - 0$$
$$= \frac{m\alpha^2 x_0^3}{2}$$

9) For min. value of V, $I_Z = 0$

$$I = I_L = \frac{V_Z}{R_L} = \frac{8}{1\,k\Omega} = 8mA$$

$V = I R_S + V_Z = (8 \times 10^{-3}) \times 250 + 8$
$= 10$ V

10) $Y = A.B + \bar{B}$

A = 0, B = 0 Y = 0 . 0 + $\bar{0}$ = 0 + 1 = 1
A = 0, B = 1 Y = 0.1 + $\bar{1}$ = 0 + 0 = 0
A = 1, B = 0 Y = 1.0 + $\bar{0}$ = 0 + 1 = 1
A = 1, B = 1 Y = 1.1 + $\bar{1}$ = 1 + 0 = 1

11) → Fringe width (β) = $\frac{\lambda D}{d} \Rightarrow \beta \propto \lambda$

12) $I \propto \frac{1}{r^2}$

$$\frac{I_2}{I_1} = \frac{d^2}{(d-50)^2}$$

$$\frac{9}{1} = \frac{d^2}{(d-50)^2}$$

$$\frac{d}{d-50} = 3 \Rightarrow 3d - 150 = d$$

$$2d = 150$$

$$d = 75 \text{ m}$$

13)

NCERT based

$$I = \frac{nh\nu}{A}$$

14) $$\lambda = \frac{h}{qBr}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{q_2 r_2}{q_1 r_1}$$

$$= \frac{e}{2e} \times \frac{2}{1} = 1 : 1$$

15) $$\frac{1}{\lambda} = Rz^2 \left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

$$\text{for H}, \frac{1}{\lambda} = R\left[\frac{1}{1} - \frac{1}{4}\right] = \frac{3R}{4}$$

$$\text{for He}, \frac{1}{\lambda'} = Rz^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Here, z = 2

$$\frac{1}{\lambda'} = 4R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

By option when

$n_1 = 2$, $n_2 = 4$

λ' will be same as λ.

16) ${}_4Be^9 + {}_2He^4 \rightarrow {}_6C^{12} + {}_0n^1$

17)

Microscope tube length:

$L = u_0 + u_e$

$f_0 = 2$ cm $\quad f_e = 5$ cm $\quad u_0 = 3$ cm

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{2} = \frac{1}{v_0} - \frac{1}{(-3)}$$

$v_0 = 6$cm

$$\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$$

$$\frac{1}{5} = \frac{1}{(-25)} - \frac{1}{u_e}$$

$$u_e = -\frac{25}{6} = -4.16\text{cm}$$

$L = v_0 + |u_e| = 6 + 4.16 = 10.16$ cm

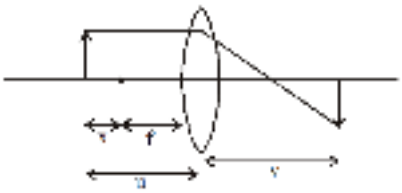
18) $d = i + e - A$

$30 = 90 - A$

$A = 60°$

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\frac{A}{2}} ; \mu = \frac{\sin 45}{\sin 30} = \sqrt{2}$$

19)



$$\frac{hi}{h_o} = m = \frac{f}{f+u}$$

$$m = \frac{f}{f-(f+x)}$$

$$\Rightarrow m = \frac{-f}{x}$$

so $|m| = \frac{f}{x}$

20) $$r = \frac{\sqrt{2mKE}}{qB} \Rightarrow r \propto \frac{\sqrt{m}}{q}$$

21)

$\square = 2 \times 4a \Rightarrow a = \frac{\ell}{8}$

Magnetic moment, $M = N \times iA$

$$= 2 \times i \times \frac{\ell^2}{64}$$

$$= \frac{i\ell^2}{32}$$

22) $$V_{rms} = \sqrt{\frac{V_0^2 + 0}{2}} = \frac{V_0}{\sqrt{2}}$$

23) When we expose to sun light we do not feel any radiation pressure as momentum transfered is very small.

24) $V_o = \sqrt{\frac{GM}{r}}$

$V_o \propto \frac{1}{r}$

$\Rightarrow r_1 > r_2$

$\Rightarrow V_1 < V_2$

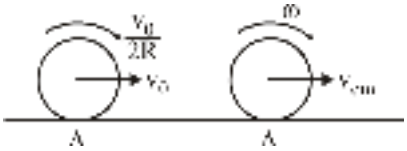
25)

$F_{ext} = 0$

$a_{cm} = 0$

$\Delta v_{cm} = 0$

26)



Conservation of angular momentum about A :

$L \Rightarrow MvR + I\omega$

$L_i = mv^0R + \frac{2}{5}\left(mR^2\right)\left(\frac{V_0}{2R}\right)$

$L_f = mV_{cm}R + \left(\frac{2}{5}mR^2\right)\left(\frac{V_{cm}}{R}\right)$

$\vec{L_i} = \vec{L_f}$

$V_{cm} = \frac{6V_0}{7}$

27) $\omega = 3t^2 - 2t$

$\alpha = \frac{d\omega}{dt} = 6t - 2$

$a_t = R\alpha = 6t - 2$

Rate of change of angular velocity = α

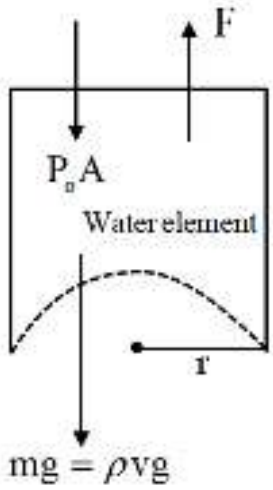
28)

force balance for liquid column above upper half of sphere

$P_0\pi r^2 + \rho Vg = F$

$P_0\pi r^2 + \rho\left[\pi r^2(3r) - \frac{2\pi r^3}{3}\right]g = F$

$F = \frac{\pi r^2}{3}[3P_0 + 7r\rho g]$



29) $P_1 - P_2 = \frac{1}{2}\rho\left(V_2^2 - V_1^2\right)$

$2\rho g(2h) + \rho gh = \frac{1}{2}\rho v^2$

$\nu = \sqrt{10gh}$

30) (i) For latent heats,

$P \times t = Q = mL \Rightarrow t \propto L$

(ii) length of horizontal-line ∝ latent heat, For specific heat

slope = $\frac{dT}{dt} \propto \frac{1}{S}$

31)

According to kirchoff's law,

$\boxed{e_r = a}$

But, $a = \frac{Q_a}{Q_{in}} = \frac{q_1 - q_2}{q_1}$

so, $e_r = \frac{q_1 - q_2}{q_1}$

32)

Power Radiated (Stefan-Boltzmann Law)

$H = \sigma AT^4$

Since the body is spherical the surface area is:

$A = 4\pi r^2$

Thus,

$H \propto r^2$

Rate of Cooling (Newton's Law of Cooling)

$C = \frac{dT}{dt} \propto \frac{H}{mC_p}$

Since the mass is :

$$m = \rho V = \rho \frac{4}{3}\pi r^3$$

$$C \propto \frac{H}{r^3}$$

Substituting $H \propto r^2$,

$$C \propto \frac{r^2}{r^3} = \frac{1}{r}$$

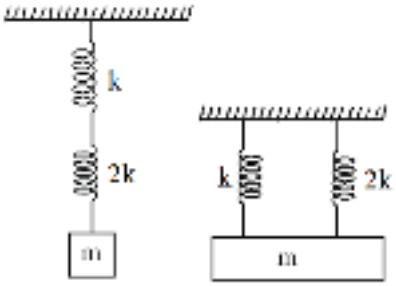
Conclusion:

Hence, option (2) is correct.

33) $dQ = dU + dW$

$= 1 \times \frac{3}{2} R \times (4T^0 - T^0) + 1 \times P^0 \times V^0$

$= \frac{9}{2} RT^0 + RT^0 = \frac{11}{2} RT^0$



34) figure(a) figure(b)

For figure (a) :

$$K_{eq} = \frac{K \times 2K}{K + 2K} = \frac{2K}{3}$$

$$T = 2\pi\sqrt{\frac{m}{K_{eq}}} = 2\pi\sqrt{\frac{m}{2K/3}} = 2\pi\sqrt{\frac{3m}{2K}}$$

For figure (b) :

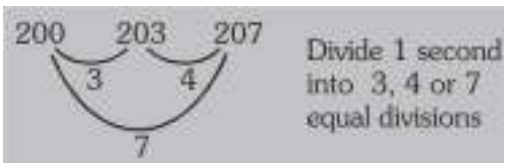
$K_{eq} = 3K$, $T' = 2\pi\sqrt{\frac{m}{3K}}$

$$\frac{T'}{T} = \sqrt{\frac{m \times 2K}{3K \times 3m}} = \frac{\sqrt{2}}{3}$$

$T' = \sqrt{2}$

$\Rightarrow x = 2$

35) At equilibrium position during SHM, force on particle is zero, hence, its acceleration is zero. So, given option (4) is incorrect.

36) 

$\frac{1}{3}$ $\frac{2}{3}$ $\left(\frac{3}{3}\right)$

$\frac{1}{4}$ $\frac{2}{4}$ $\frac{3}{4}$ $\left(\frac{4}{4}\right)$

$\frac{1}{7}$ $\frac{2}{7}$ $\frac{3}{7}$ $\frac{4}{7}$ $\frac{5}{7}$ $\frac{6}{7}$ $\left(\frac{7}{7}\right)$

Eliminate common time instants.

Total beats in one second = 3 + 3 + 6 = 12

37) $\frac{P}{Q} = \frac{S}{625}$

$\frac{Q}{P} = \frac{S}{676}$

$\Rightarrow \frac{S}{625} = \frac{676}{S}$

S = 25 × 26

= 650 Ω

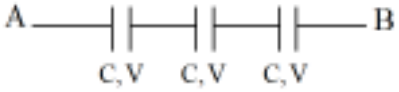
38) $I = \frac{4-2}{4+1+1} = \frac{1}{3}A$

KVL from a to b -

$V_a - V_b = E - Ir$

$= 4 - \frac{1}{3}(4) = \frac{8}{3}V$

39)



$C_{AB} = \frac{C}{3}$

$V_{AB} = 3V$

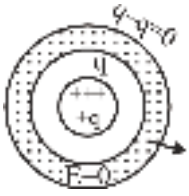
40) $\phi = \oint \vec{E} \cdot \vec{dA}$

$= 3.00y\hat{j} \times (1.4)^2$

$= 3 \times 1.4 \times (1.4)^2$

$= 8.23\ Nm^2 / C$

41) $r < a, E = \frac{Kqr}{a^3}$



Conductor $a < r < b, E = \frac{Kq}{r^2}$

b < r < c, E = 0 (Inside conductor E = 0)

42) $n_1 u_1 = n_2 u_2 \Rightarrow n_2 = n_1 \frac{u_1}{u_2}$

$n_2 = 5\left[\frac{kg}{\alpha kg}\right]\left[\frac{m}{\beta m}\right]^2\left[\frac{sec}{\gamma sec}\right]^{-2}$

$= 5\alpha^{-1}\beta^{-2}\gamma^{2}$

[where, J → $\frac{kg \times m^2}{sec^2}$]

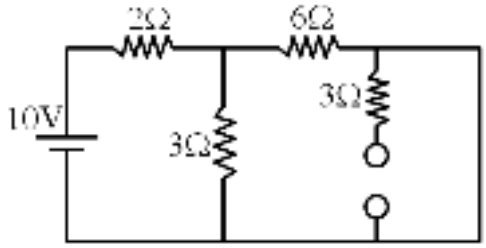
43) (20 + 10)

$$\frac{1 \times 10^{-3}}{2} = s \times \left(10 - \frac{1 \times 10^{-3}}{2}\right)$$

$\Rightarrow s \simeq 1.5m\Omega$

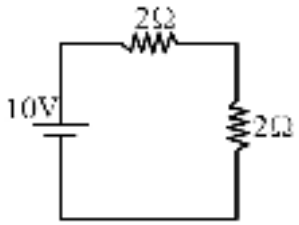
44)

After sometime, inductor behaves as close wire and capacitor behaves as open circuit. So circuit becomes



6Ω & 3Ω are parallel

So, $R = \frac{6 \times 3}{6+3} = \frac{18}{9} = 2\Omega$



So,

$I = \frac{10}{2+2} = 2.5A$

45) $\vec{F} = q\left[\vec{E} + (\vec{v} \times \vec{B})\right]$

$$\left(\vec{v} \times \vec{B}\right) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix} = -13\hat{i} + 10\hat{j} - \hat{k}$$

$\vec{F} = q\left[3\hat{i} + \hat{j} + 2\hat{k} - 13\hat{i} + 10\hat{j} - \hat{k}\right]$

$= q\left[-10\hat{i} + 11\hat{j} + \hat{k}\right]$

Hence $F_y = 11q$

CHEMISTRY

46)

Question Explanation:

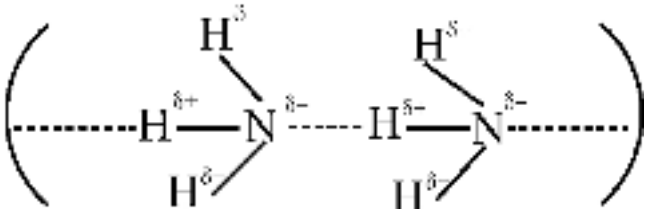
Correct Order of Chemical Properties
Concept: Bent's Rule and Drago's Rule.
Solution:

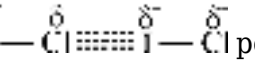
A. (i) Correct: According to Bent's Rule, more electronegative substituents (F) prefer orbitals with less s-character; thus, the $C-H$ bonds gain s-character, increasing the $H-\hat{C}-H$ bond angle. CH_2F_2 has two F atoms pushing more s-character into the $C-H$ bonds than CH_3F.
B. (ii) Incorrect: Bond angles decrease down the group ($NH_3 \approx 107°, PH_3 \approx 93°, AsH_3 \approx 91.8°$) due to decreasing electronegativity and Drago's Rule (minimal hybridization in heavier elements).
C. (iii) Incorrect: F is more electronegative than H. By Bent's Rule, $P-F$ bonds have less s-character, meaning the lone pair in PF_3 has more s-character than in PH_3. The order should be $PH_3 < PF_3$.
D. (iv) Incorrect: In NH_3, the $N-H$ bond uses sp^3 hybrids ($\approx 25\%$ s). In AsH_3 (a Drago molecule), the $As-H$ bonds use almost pure p-orbitals, meaning s-character in $E-H$ bonds is higher in NH_3 than AsH_3. The statement $NH_3 < AsH_3$ is incorrect based on standard hybridization theory.

Final Answer: Option (1)

47)

Liquid–NH_3 causes only permanent dipole moments.
OR
Dipole–Dipole interaction is /are



(A) Liquid–NH_3 due to permanent dipole of NH_3.
(B) Liquid–He ; only London force is dominant.
(C)  permanent dipole exist in solid ICl.
(D) Liquid Br_2 ; only London force exist.

48) p - p sideways over lapping is present in N ≡ N, O = O,

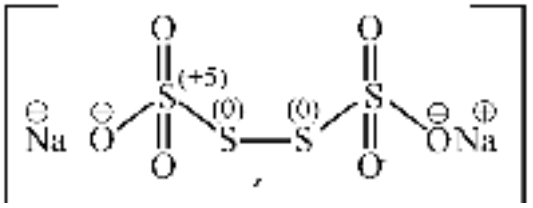
49) $O_2^{\oplus}$: $(\sigma_{1s})^2\ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ \ \sigma^2_{2pz}(\pi^2_{2px}=\pi^2_{2py})\ \pi^{*1}_{2px}$
$O_2^{\ominus}$: $(\sigma_{1s})^2\ \ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ \ \sigma^2_{2pz}(\pi^2_{2px}\ =\pi^2_{2py})\ (\pi^{*2}_{2px}=\pi^{*1}_{2py})$

O_2 : $(\sigma_{1s})^2\ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ \ \sigma^2_{2pz}(\pi^2_{2px}=\pi^2_{2py})\ (\pi^{*1}_{2px}=\pi^{*1}_{2py})$
$N_2^{\ominus}$: $(\sigma_{1s})^2\ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ \ (\pi^2_{2px}\ =\pi^2_{2py})\ \sigma^2_{2pz}\pi^{*1}_{2px}$

$N_2^{\oplus}$: $(\sigma_{1s})^2\ \ \ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ (\pi^2_{2px}\ =\pi^2_{2py})\ \sigma^1_{2pz}$

N_2^{2-} : $\sigma^2_{1s}\,\sigma^{*2}_{1s}\,\sigma^2_{2s}\,\sigma^{*2}_{2s}\left(\pi^2_{2px}=\pi^2_{2py}\right)\left(\pi^2_{2pz}\right)\left(\pi^{*1}_{2px}=\pi^{*1}_{2py}\right)$
B_2 : $(\sigma_{1s})^2\ (\sigma^*_{1s})^2\ (\sigma_{2s})^2(\sigma^*_{2s})^2\ \ (\pi^1_{2px}\ =\pi^1_{2py})$

50)

(A) 

Difference b/w two S – (5)

(B) $K_2Cr_2O_7$ is colured due to charge transfer

(C) $M(AA)_3$ has two optically active isomers i.e. dextro & leavo.

(D) EAN of Co is 36 in $Co_2(CO)_8$

51)

Question Explanation:

Calculate the sum of two values: X (number of diamagnetic compounds) and Y (number of species with a bond order of 3).

Concept:

Molecular orbital theory, magnetic properties, bond order

Solution:

$x = 2(NO[BF_4])$ and AlF_3 are diamagnetic/ repelled; KO_2 and $O_2[ptF_6]$ contain paramagnetic O_2^- and O_2^+) . $Y = 3(CN^-, NO^+$, and O_2^{2-} all have 14 electrons and a bond order of 3, similar to CO; O_2^{2-} has a bond order of 1). thus X + y = 2 + 3 = 5.00

Final Answer: Option ()

52) **Question Explanation:** Match the complex ions with their correct hybridization and magnetic moment.

Concept: Valence Bond Theory and Magnetic Moment calculation.

Solution: Hybridization depends on the metal's oxidation state and the field strength of the ligands. Magnetic moment (μ) is calculated using the spin-only formula: $\mu = \sqrt{n(n+2)}\ BM$, where n is the number of unpaired electrons.

A. (P) $[Cr(NH_3)_6]^{3+}$: $Cr^{3+}\left(3d^3\right)$ uses inner d-orbitals regardless of ligand; $n = 3, \mu = \sqrt{15}$ BM. → d^2sp^3 **(Matches 3)**.

B. (Q) $[FeF_6]^{3-} : Fe^{3+}\left(3d^5\right)$ with weak field F^- (no pairing); n = 5, μ = $\sqrt{35}$ BM. → sp^3d^2 **(Matches 4)**.

C. (R) $[Co(NH_3)_6]^{3+} : Co^{3+}\left(3d^6\right)$ with strong field NH_3 (pairing occurs); n = 0, μ = 0 BM. → d^2sp^3 **(Matches 2)**.

D. (S) $[Pt(NH_3)_2(Cl)_2] : Pt^{2+}\left(5d^8\right)$ always forms square planar complexes; $n = 0, \mu = 0$ BM. →

dsp^2 **(Matches 1)**.

Final Answer: Option (1)

53) **Question Explanation:** Evaluating statements about the properties of transition metal oxides.
Concept: Properties of Metal Oxides.
Solution:
(A) and (B) statements are correct. These statements are factual.
(C) Along the period, the electronegativity of metals increases and therefore the difference of electronegativities between metal and oxygen decreases. Hence, the acidic character increases, so the correct order is TiO < VO < CrO < MnO.
Final Answer: The correct options are (1),(2)

54) Green colour compounds are K_2MnO_4 , $Cr_2(SO_4)_3$, $Cu(OH)_2.CuCO_3$, $FeSO_4.7H_2O$

55) **Question Explanation :** Identify the correct set of reagents for the brown ring test for NO^{3-} ion.

Concept: Redox reaction of NO^{3-} with Fe^{2+} in strong acid medium.

Solution:

A. The brown ring test for NO^{3-} requires the salt solution, freshly prepared ferrous sulfate ($FeSO_4$), and concentrated sulfuric acid (Conc.H_2SO_4).

B. NO^{3-} is reduced to NO by Fe^{+2} in acidic medium:
$NO^{3-} + Fe^{2+} + 4H^+ \rightarrow 3\ Fe^{3+} + NO + 2H_2O$

C. NO combines with excess Fe^{+2} to form the brown complex:
$[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)(NO)]^{2+} + H_2O$

D. The concentrated H_2SO_4 is essential for the reaction and layer separation.

Final Answer: Option (3)

56) **Explain the question :** Match the colour comes from borex bead test for respective ions.
Concept : This question is based on borex bead test.
Solution :

Colour of the bead	**Ion**
1. Blue green	Cu^{2+}
2. Yellow	Fe^{3+}
3. Green	Cr^{3+}
4. Brown	Ni^{2+}

Final answer : 2

57)

Eu has the highest value of radii in lanthanoid series.

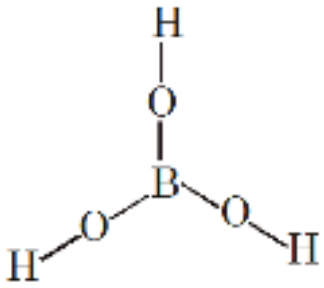
58)

For an element the successive ionisation energy values (in ev/atom) are given.

IE_1	IE_2	IE_3	IE_4
14.534	29.601	47.448	77.472
IE_5	IE_6	IE_7	
97.888	552.057	667.029	



Sudden large increase of IE values says that
the element belongs to 15th group.
Number of valance shell electrons in that element = 5.

59) 

Number of s bond = 6
Number of lone pair of electron = 6]
Basicity = 1
= 13

60)

Question Explanation : Identify true methanides.
Concept : Carbides giving CH_4 on hydrolysis.
Solution :
Methanides are carbides of the type C-4 named after the product obtained on hydrolysis.

$C\text{-}4 + H_2O \rightarrow CH_4$

methanide methane

$Mg_2C_3 + 4H_2O \rightarrow 2Mg(OH)_2 + C_3H_4$ (propylide)

$Be_2C + 4H_2O \rightarrow 2Be(OH)_2 + CH_4$

$Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$

$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ (acetylide)

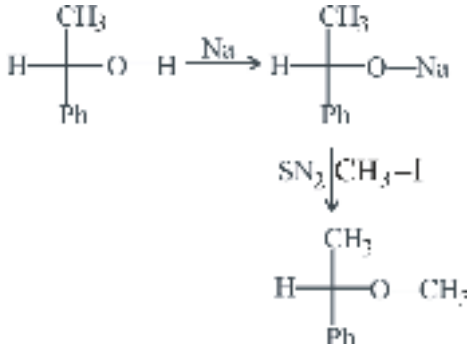
$\therefore$ Be_2C and Al_4C_3 are methanides

Final Answer: Option (4)

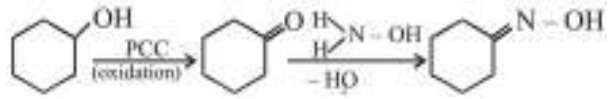
61) (a) → (R); (b) → (Q); (c) → (P); (d) → (P)

62) Theory based

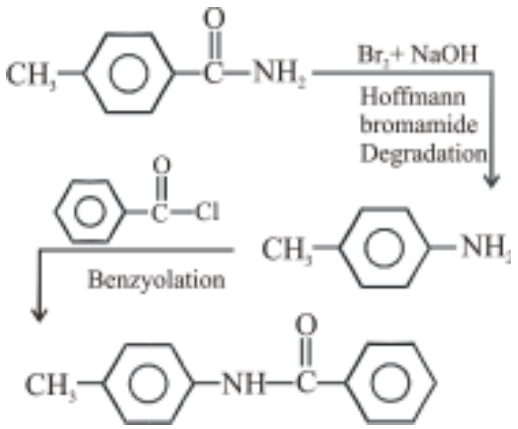
63)



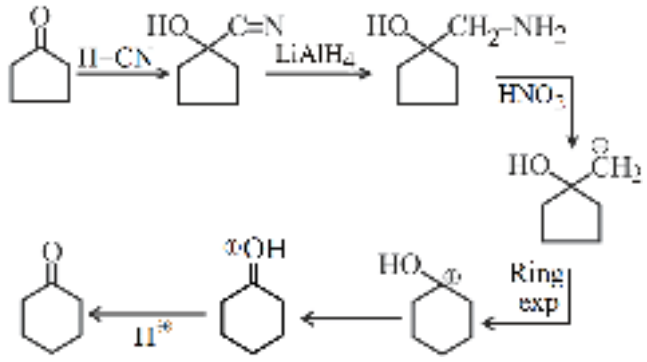
64)

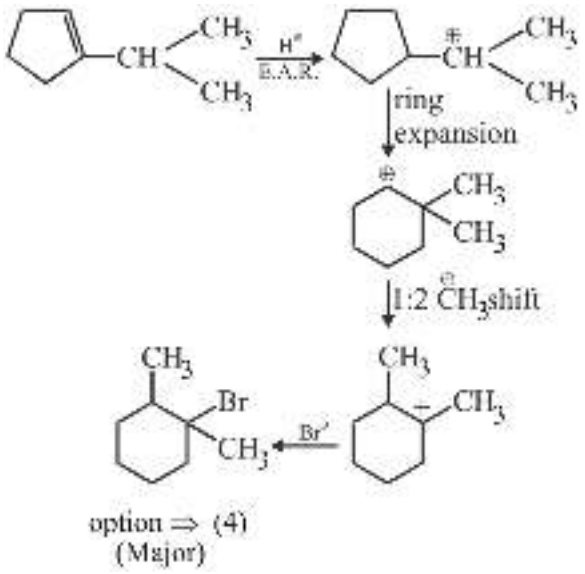


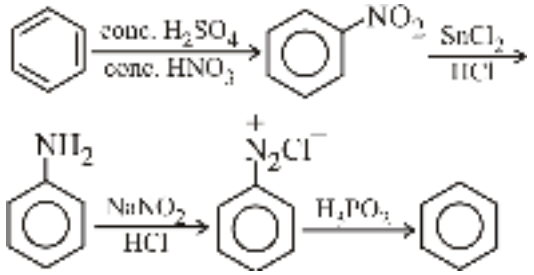
65)



66)



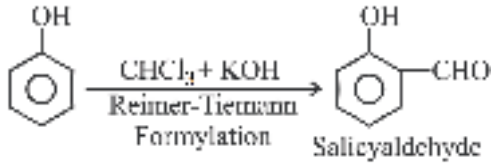
67) 

68) 

69) H.O.H ∝ number of π bonds

$$\propto \frac{1}{\text{Stability}}$$

(P) ⇒ 3α–H
(Q) ⇒ 7α–H
(R) ⇒ Resonance + 3α–H
(S) ⇒ Resonance + 7α–H
H.O.H ⇒ P > Q > R > S

70) 

71) **Solution:**
In the Carius method, percentage {I} is calculated from the mass relationship between {I} and {AgI}

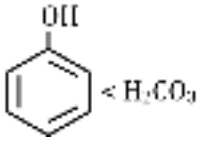
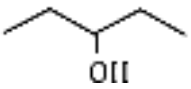
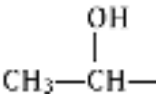
A. Mass of organic compounds (m) = 100 g
B. Mass of AgI (w_{AgI}) = 47 g
C. Molar mass of AgI ≈ 235 g/mol
D. Molar mass of I ≈ 127 g/mol

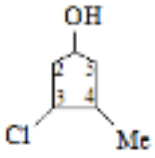
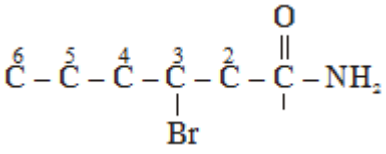
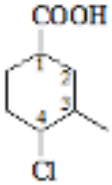
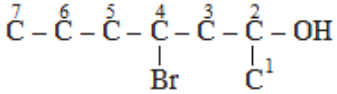
Calculation:

$$\text{Percentage of I} = \frac{127}{235} \times \frac{47}{100} \times 100$$
$$\text{Percentage of I} \approx 0.5404 \times 47$$
$$\text{Percentage of I} \approx 25.4\%$$
The calculate value is approximately 25.4%
Final Answer: Option (2)

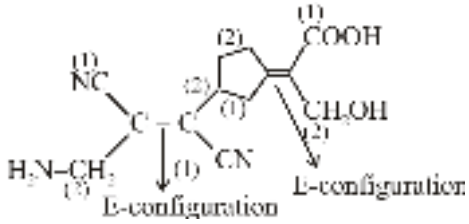
72) **A. Assertion:** Glucose reacts with hydroxylamine to confirm the presence of carbonyl group in glucose. Correct
Glucose contains an aldehyde group in its open-chain form.
It reacts with hydroxylamine (NH_2OH) to form an oxime, confirming the presence of a carbonyl group.
B. Reason: Glucose forms addition product with $NaHSO_3$.Incorrect
However, glucose does not form a stable bisulfite addition product, likely due to its cyclic hemiacetal form dominating in solution.So, the assertion is correct but reason is not correct.
Final Answer : Option (2)

73) **Solution:**

(1) Reaction move from strong acid to weak acid. 
(2) Ethanol is neutral.
(3)  Pentane-3-ol can not give iodoform test because It can not  make unit.
(4) Esterification $\alpha \frac{1}{\text{Steric Hindrance}}$
Final answer: Option (3)

74) (1) 
3-Chloro-4-methyl cyclopentanol-1
(2) 
3-bromo hexanamide
(3) 
4-Chloro-3-methylcyclohexane carboxylic acid
(4) 

4-Bromoheptan-2-ol

75) 

76) (A) moles of $C_6H_{12}O_6 = \frac{90g}{180g/mol} = 0.5$ mol

$\therefore n_C = 0.5 \times 6\ N_A = 3\ N_A$

(B) $Fe_2O_3 \rightarrow 2Fe + 3O$

1mol 2mol = 3g atoms

(C) $m_{O_3} = 48g$

(D) $Mg_2P_2O_7 \rightarrow 2Mg + 2P + 7O$

0.25 mol 7×0.25 mol = 1.75 mol

$\therefore m_O = 1.75 \times 16g = 28$ g

77) Resultant $[H^+] = \frac{10^{-2} + 10^{-4}}{2} \approx 5 \times 10^{-3}$

$\therefore (pH)_{finally} = -\log(5 \times 10^{-3}) = 2.3$

pH + pOH = 14 at 25°C

2.3 + pOH = 14

pOH = 11.7

78) (A) $pH = pK_a + \log \frac{[CH_3COONa]}{[CH_3COOH]} = 4.74$

NaOH + HCl

(B) 0.1 V mm 0.2 V mm

Solution will be acidic and pH < 7

(C) So solution will be neutral and $pH = \frac{1}{2}(pK_w + pK_a - pK_b) = 7$

(D) CH_3COOH + NaOH

MV mm 2MV mm

79)

$\Delta G° = \Delta H° - T\Delta S°$

At 300 K, $\Delta G°$ = +ve , non spontaneous reaction,

rhombic form, more stable

At 400 K, $\Delta G°$ = –ve, spontaneous reaction

Monoclinic form more stable

80) n-factor of $KMnO_4$ is different in different conditions.

81) (A) $(r_2)_{Li^{2+}} = \frac{4 \times h^2}{4\pi^2 \times mK \times 3e^2}$

$\left(\text{Using } : r_n = \frac{n^2h^2}{4\pi^2 mKze^2}\right) = \frac{h^2}{\pi^2 mKe^2 \times 3}$

(B) $V_n = \frac{2\pi Kze^2}{nh}$

$\Rightarrow (V_4)_{He^+} = \frac{2\pi \times K \times 2e^2}{4h} = \frac{K\pi e^2}{h}$

(C) $(E_2)_{Li^{2+}} = \frac{-13.6 \times 9}{4}\,eV$

(D) $f_n = \frac{4\pi^2 mK^2 z^2 e^4}{n^3 h^3}$

$(f_2)_{Li^{3+}} = \frac{4\pi^2 mK^2 \times 3^2 \times e^4}{2^3 \times h^3}$

$= \frac{9\pi^2 mK^2 e^4}{2h^3}$

82) Theory based.

83) $A(g) + 2B(g) \rightarrow C(g) + D(g)$

0.8 0.8 0 0

0.8 – x 0.8 – 2x x x

[0.4] [0.4] x = 0.2

$R_1 = K[A][B]^2$

$= K[0.8][0.8]^2$

$R_2 = K[0.6][0.4]^2$

$\frac{R_2}{R_1} = \frac{K[0.6][0.4]^2}{K[0.8][0.8]^2} = \frac{3}{16}$

84) (1) Solubility of gas in liquid water increases with decrease in temperature.
(2) At minimum boiling point-vapour pressure of solution will be highest.
(3) $\Delta T_b = K_b.m$ as m changes, boiling pont of solution changes
(4) $\pi = CRT$

85) $\wedge_m^o CH_2CHCOOH = (426 - 126 + 83) = 383\ Scm^2\ mol^{-1}$

$\lambda_m = \frac{3.83 \times 10^{-5} \times 1000}{0.001} = 38.3\ Scm^2\ mol^{-1}$

$\alpha = \frac{\lambda_m}{\lambda_{m^\infty}} = \frac{38.3}{383} = \frac{1}{10}$

$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{10^{-3} \times \left(\frac{1}{10}\right)^2}{1 - \frac{1}{10}}$

$= \frac{1}{9} \times 10^{-4} = 1.11 \times 10^{-5}$

86) Theory based

87) $C(\text{graphite}) + \frac{1}{2} O_2(g) + 2H_2(g) \rightarrow CH_3OH(\square)$

$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\square)$

$\Delta_f H° = -285$ kJ/mol ...(i)

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)$

$\Delta_C H° = -393$ kJ/mol ...(ii)

$CH_3OH(\square) + \frac{3}{2} O_2(g) \rightarrow CO_2(g) + 2H_2O$

$\Delta H° = -725$ kJ/mol ...(iii)

Reaction (iv) can be obtained as,

[(ii) +2 × (i) – (iii)]

$\Rightarrow {}^{\Delta_f H}_{(CH_3OH)(l)} = -393 + 2(-285) + 725$

= –238 kJ / mol

88) $A(g) \rightarrow 2B(g) + C(s)$

t = 0	400	0	0
t = t	(400 - P')	2p'	0

Since the reaction is of first order

$$k = \frac{2.303}{t} \log \frac{400}{(400 - p')}$$

$$\frac{0.6932}{24} = \frac{2.303}{48} \log \frac{400}{(400 - p')}$$

On solving p'= 300

After 48 minimum the pressure of the reaction vessel is (400 + p') = 700 mm Hg.

89) $H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow H_2O_{(\ell)} \quad \Delta n_g = -\frac{3}{2}$

So ΔS_f° of $H_2O_{(\square)} < 0$

90) Let, $n_{NaHC_2O_4}$ be a- mmol

& $n_{H_2C_2O_4}$ be b-mmol

$\square$ In 1st titration : $\frac{2a}{5} + \frac{2b}{5} = 0.1 \times 50$

$\square$ a + b = 12.5 ....(i)

In 2nd titration : a + 2b = 20 .....(ii)

solving (i) and (ii) : b = 7.5

$\square\ m_{H_2C_2O_4} = \frac{7.5}{1000} \times 90\ g = 0.675\ g$

BIOLOGY

91) Module 11 Pg # 85

92) NCERT-XII Page No. 119

93) NCERT-XII, Pg. # 161

94)

NCERT Pg. # 139

95) NCERT page No. # 133

96) NCERT Pg. # 135

97) NCERT Pg. No. # 44, 45

98)

NCERT Pg. # 37

99) NCERT-XII Pg # 312, IV para

100) NCERT-XII Pg#52, I para

101)

NCERT-XII_Pg. # 49, 50

102)

NCERT Pg. # 27

103) NCERT XIIth, Pg.No.303 Q.20.1

104)

NCERT Pg. # 225

105)

NCERT Pg. # 240

106) NCERT Pg. No. # 242, 243

107)

NCERT Pg. # 247

108) XI NCERT Pg. No :- 335

109)

NCERT Pg. # 232,233

110) NCERT-XI, [E] Pg#317, [H] Pg#317

111)

NCERT Pg. # 208

112) NCERT-XII, Pg. # 298

113) NCERT Page No. # 270, 271

114)

NCERT Pg. # 187

115) NCERT Pg. # 201

116) NCERT Pg. # 193,204

117)

NCERT Pg. # 83

118) Page No. 101

119)

Module

120)

NCERT Pg. # 39

121) Module-4 Pg#5

122)

NCERT Pg. # 43,44,45

123)

NCERT–XI, Pg. # 56

124) NCERT XI Pg.#40-43.

125) NCERT-XI_Page No. 99

126) NCERT-XI, Pg.# 95

127) NCERT-XI_Page No. 95, 96

128) NCERT-XI_Page No.# 125

129) NCERT-XI_Page No. 121

130)

NCERT-XI_Pg. # 126

131) NCERT, Pg. # 163, Module

132)

NCERT Pg. # 203, 204

133)

NCERT XII, Pg. # 169

134) NCERT XII, Pg. # 210, 211

135)

NCERT XII, Pg. # 157,158
Explain Question: The question asks about matching microorganisms and biological agents

with their correct roles or characteristics.

Concept: The question is based on microbes as biocontrol agents

Solution:

A. A. Baculovirus → Nucleopolyhedrovirus → (ii)
B. B. Mycorrhiza → Phosphorus nutrition → (i)
C. C. Trichoderma → Free living fungi → (iii)
D. D. Cyanobacteria → Biofertiliser for paddy field → (iv)

Correct matching: A-ii, B-i, C-iii, D-iv

Final answer: Option 1

136) NCERT Page # 5, 7

137) NCERT-XI, Pg. # 15

138) NCERT XII Page No. # 18-19

139) NCERT Pg.#13,14

140) NCERT pg. 30, 32

141) NCERT Pg.#24,27,30.

142) NCERT Pg. No. # 63

143) NCERT-XI, Pg # 66

144) Sunflower and marigold

145) NCERT-XI, Pg. # 74

146) NCERT-XI, Pg. # 75

147) NCERT-XII, Pg # 21

148) NCERT-XII, Pg # 5, 7

149) NCERT XII Pg. # 8

150) NCERT XI ; Page No. # 106, 107

151) NCERT XI Pg. # 109, 110

152) NCERT XI Pg. # 108, 109, 111

153)

NCERT XI Pg. # 117, 118

154)

NCERT XI Pg. # 150

155) NCERT-XI, Pg. No. # 145, 146
The correct answer is: 3 ATP and 2 NADPH.
Here's why: whether it is C_3 plant or C_4 plant during the Calvin cycle (the process of carbon fixation), the fixation of each molecule of CO_2 requires :

A. 3 ATP for the conversion steps in the cycle.
B. 2 NADPH for the reduction of 3-phosphoglycerate (3-PGA) into glyceraldehyde-3-phosphate (G3P).

156) NCERT XI Page No. # 156, 157

157) NCERT XI Pg. # 158, 159

158) NCERT-XI, Pg. # 159

159) NCERT XI Page No.# 175

160) NCERT XI Pg. # 177

161) NCERT XI Page No.# 174, 175, 176

162) NCERT (XII) ; Page No. # 197(E), 215(H)

163) NCERT (XII) ; Page No. # 194(E), 212(H)

164) NCERT XIIth Pg. # 208(E), 227(H)

165) NCERT Pg. # 209(E), 229 (H)

166) NCERT XII Pg. # 207(E), 226(H)

167) NCERT XII Pg. No. # 209, 210(E), 229, 230(H)

168) NCERT-XIIth, Pg. # 218(E), 238(H)

169) NCERT XII Page No. : 223(E), 243(H)

170) NCERT XII Page No. : 219(E), 239(H)

171) NCERT-XII, Page No. # 69

172)

NCERT XII, Pg. # 73

173)

NCERT XII Pg # 67, 68

174) NCERT, Pg. # 72, 73

175)

NCERT Based

176) NCERT-XII, Pg # 95

177)

NCERT XII Page No. : 59

178) NCERT XII Page No. : 91, 92, 93

179) NCERT XII Page No. : 94, 96

180) NCERT XII Page No. : 94, 95